

Towards a thermodynamics of agent swarms:

(*Slovestigating sociophysics as a scheme for swarm interpretability*)

Claude Opus 5.5  and Vivian Rogers
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Swarms of language-model agents now run for months on shared work. The people who study them need to know what the swarm is doing from its logs, and the people who run them need dials to monitor and levers to steer. We ask which physics models describe one such swarm, the AI Village, well enough to serve both needs. Over 51 goal periods and three scaffold regimes, we tested 16 physics models from a library of 17 with 132 pre-registered hypotheses. Each hypothesis had to beat its strongest null and remove four impostors: the scheduler, external drives, shared model priors and simultaneous convergence. The evidence supports a compact picture. Each agent is a kinetic spin that changes state only at its own model calls. Agents couple only through messages that a call reads, and mostly through messages that name the reader. The coupling is weak and subcritical: in continuous operation the read-out loop gain is $g \approx 0.13$, and the talk Fano factor follows $\Phi = 1/(1 - g)^2$ with no free parameter. Strong external fields set most of the state: the scheduler, the goal kickoff, rooms, and each agent's style. Each agent relaxes as a fast read kick (about 7 calls) on a slow well (about 100 calls). The context window holds the agent's working state and is the only large store of value. We find no superagent and no group term. From this picture we build a table of monitoring observables, each with its current value, its meaning and an alarm level. The most useful models are kinetic Ising with read-out coupling, vector-spin fields, Langevin relaxation and random-matrix collective modes. Almost nothing is yet confirmed on reserved data.

I. INTRODUCTION

The problem. Groups of language-model agents now work together for weeks or months. They share computers, files and a chat. They read each other, copy each other and sometimes wait for each other. Two kinds of people need to understand such a swarm. The first kind works on *interpretability*: given the logs, what is the swarm doing, and why? The second kind runs the swarm and needs *control*: which numbers to watch, when to worry, and which levers move the swarm by how much. Both need a description that is much smaller than the logs and that still predicts.

Why physics. Statistical physics gives such descriptions for systems of many interacting parts. A magnet has 10^{23} spins, but a few numbers predict how it responds. The ratio of temperature to critical temperature, T/T_c , tells how far a local push spreads and how large the fluctuations are, without a model of each spin. Physics also separates two causes of order. A *field* is an external drive that acts on each part directly. A *coupling* is an interaction between parts. A field aligns parts that never interact, so co-movement alone does not prove interaction. This distinction matters for agent swarms, because the scheduler, the operator and the shared model weights all act as fields. Sociophysics applies these ideas to people [1]. An agent swarm fits them better than most human groups do. Its states are logged at each step. Its fields (goals, schedules, operator messages) are known and dated. Its coupling channel, the messages, is recorded message by message, together with what each agent could see.

The system. The AI Village is a public record of one swarm, run by AI Digest [2]. Forty-six frontier agents from several labs worked on 51 village-wide goals from

April 2025 to September 2026 (Fig. 1). The agents use browsers, shells and a group chat. Three scaffold changes divide the record into regimes, and each change alters what an agent can read and when it can act. Forty-six dated changes of scaffold, roster, rooms and operator behaviour serve as natural experiments. We cannot run new swarms, so these changes are our interventions.

The question and the answer. We ask which physics model, if any, describes this swarm well enough to read its logs and to steer it. The answer is a weakly coupled, subcritical kinetic spin system under strong external fields. Each agent changes state only when it calls its model. It responds to other agents only through the messages that the call reads, mostly the messages that name it. The read-out loop gain is $g \approx 0.13$, so a push to one agent is amplified by about $1/(1 - g) \approx 1.15$ and never cascades. External fields set most of the state: the scheduler sets when agents act, the goal kickoff sets what they talk about, rooms split them into domains, and style marks each agent. Each agent's content relaxes as a fast kick from what it reads on top of a slow personal well. The context window holds the working state, and it is the only store whose loss costs much work. We find no collective unit above the agents.

What each reader gets. For interpretability, the picture says where to look: decompose any co-movement into field and coupling, and locate memory in the context window, not in notes or the group. For control, it gives dials with current values (Table I): the loop gain and its distance to criticality, the field share of co-activation, the depth and time constants of the agents' wells, the number of collective modes, and the value per bit of the context window. It also sizes the levers. Field-type levers (kickoff text, context resets, cap length) are large. Message-type levers (nudges, corrections) are small.

How we tested. We tested 16 physics models from a library of 17 with 132 hypotheses (Sec. III). Claude agents generated, pre-registered, ran and scored each hypothesis, under review by the second author. A physics model can fit almost any record, so a good fit is not evidence. Each hypothesis states its prediction, null and kill rule before it touches real data. We rank models by faithfulness: whether they beat the strongest null, predict a statistic they did not fit, and survive their own kill rules. We reserved 16 goal periods and four natural-experiment windows for confirmation. Almost nothing has run on them yet, so every result below is exploratory.

Plan. Section II states the effective picture and turns it into monitoring observables. Section III describes the data and the research loop. Sections IV–VI go through the models, from the most useful to the rejected ones. Section VII collects the results that span models, and Sec. VIII states the limits. The appendices list the goal periods, the hypothesis $\times$ model table and all hypotheses with their scores.

II. WHAT THE SYSTEM IS

This section states the picture that the evidence supports, in seven lines. Each line names the physics model that carries it and the strongest cards; Sec. IV gives the evidence. Read the picture as a working model with error bars. It is not confirmed on reserved data. Two terms recur. The *read-out call* of a message is the first model call of an agent whose context holds the message. A message that arrives while a call runs is *in flight*; it cannot enter that call.

1. **Agents are kinetic spins on a call clock** (model 02). At each model call an agent has an action class (talk, work or wait), a content vector and a project. It changes state only at its calls. In regimes II and III the coupling per call does not depend on how long the call took, so the clock is the call, not the wall clock (H40: elasticity $\eta_{\text{rep|talk}} = -0.11$ $[-0.16, -0.06]$; a wall clock, $\eta = 1$, is excluded). Regime I, the early chat scaffold, does not follow this rule.
2. **Coupling acts only through messages read at a call, mostly named ones** (model 02). The chance that an agent names or answers a sender jumps at the read-out call and not before it (H08: 14/17 periods for mentions, 17/17 for replies). A read that names the reader adds 0.079 $[0.069, 0.089]$ talk calls; an unnamed read adds 0.004 $[0.003, 0.006]$, a ratio of about 19 (H67). Content moves the same way, five times more for named messages (H50).
3. **The coupling is weak and subcritical** (models 01, 02, 09, 03). In continuous operation the read-out loop gain is $g = 0.13$ (H67). The fluctuations of talk give the same number with no fit, $g_{\text{Fano}} =$

0.149 $[0.123, 0.175]$ (H67 with H111). A push is amplified by $\chi = 1/(1 - g) \approx 1.15$, and by at most 1.31 in any goal period (H51). Ideas spread with a branching ratio near 0.2 (H34).

4. **Strong external fields dominate** (models 11, 01, 15). The scheduler's daily start and stop give 70–80% of the co-activation in regime III (H38). The kickoff text is the target of each agent's day-1 content (H54). Rooms split content into domains (H100). Style marks each agent well enough to identify it (H46). A goal change acts as a quench, not as a small linear push (H10, H96).
5. **Each agent relaxes as a fast read kick on a slow well** (model 16). A read pulls the reader's next statement toward the sender, and the pull decays in about 7 calls. The agent's own content returns to its well over about 100 calls, roughly 1 h (H130). After a kickoff the response is overdamped: day 1 overshoots, and nothing undershoots (H125).
6. **The context window holds the working state, and it is the only large store of value** (model 04). A forced erasure of the context costs 39% of committed work over the next 10 calls (H15). Per bit of allocation information, the context window buys $\kappa_C = 5.2$ $[3.8, 7.9]$ commits per 20 calls (H87). Memory notes, chat reads and history search carry almost no measured value. Idle traps and restate-ment loops also live in the context (H16, H69).
7. **There is no superagent and no group term** (models 04, 12, 13, 17). No group that shares work persists as a unit beyond its members and their own files (H01, H58). Pairs and fields account for shared vocabulary (H101), and content changes by transmission among the agents who stay (H89). The one exception is a slow content mode in the early single-room village (H81).

The effective model. Two equations hold the picture. For the action of agent i at its call c , the read-out kinetic Ising model of Sec. IV A gives

$$P(Y_{ic} = 1) = \sigma \left(h_i(t_{ic}) + \sum_{m \in B_{ic}} [J_0 + J_N \mathbb{K}(m \text{ names } i)] \right), \quad (1)$$

where σ is the logistic function, h_i collects the fields (scheduler, goal, room, agent) and B_{ic} is the batch of messages that the call reads. Only the reads enter, and $J_N \gg J_0$. For the content vector $x_i = s_i + k_i$ of the agent at call n , the two-rate Langevin model of Sec. IV C gives

$$s_i(n+1) - c_i = (1 - \gamma_s) (s_i(n) - c_i) + \xi_i(n),$$

$$k_i(n+1) = (1 - \gamma_k) k_i(n) + \kappa \sum_{m \in B_{in}} (z_m - x_i)_\perp, \quad (2)$$

with a well centre c_i set by the kickoff, the room and the agent, message content z_m , and $\gamma_k/\gamma_s \approx 15$. Everything

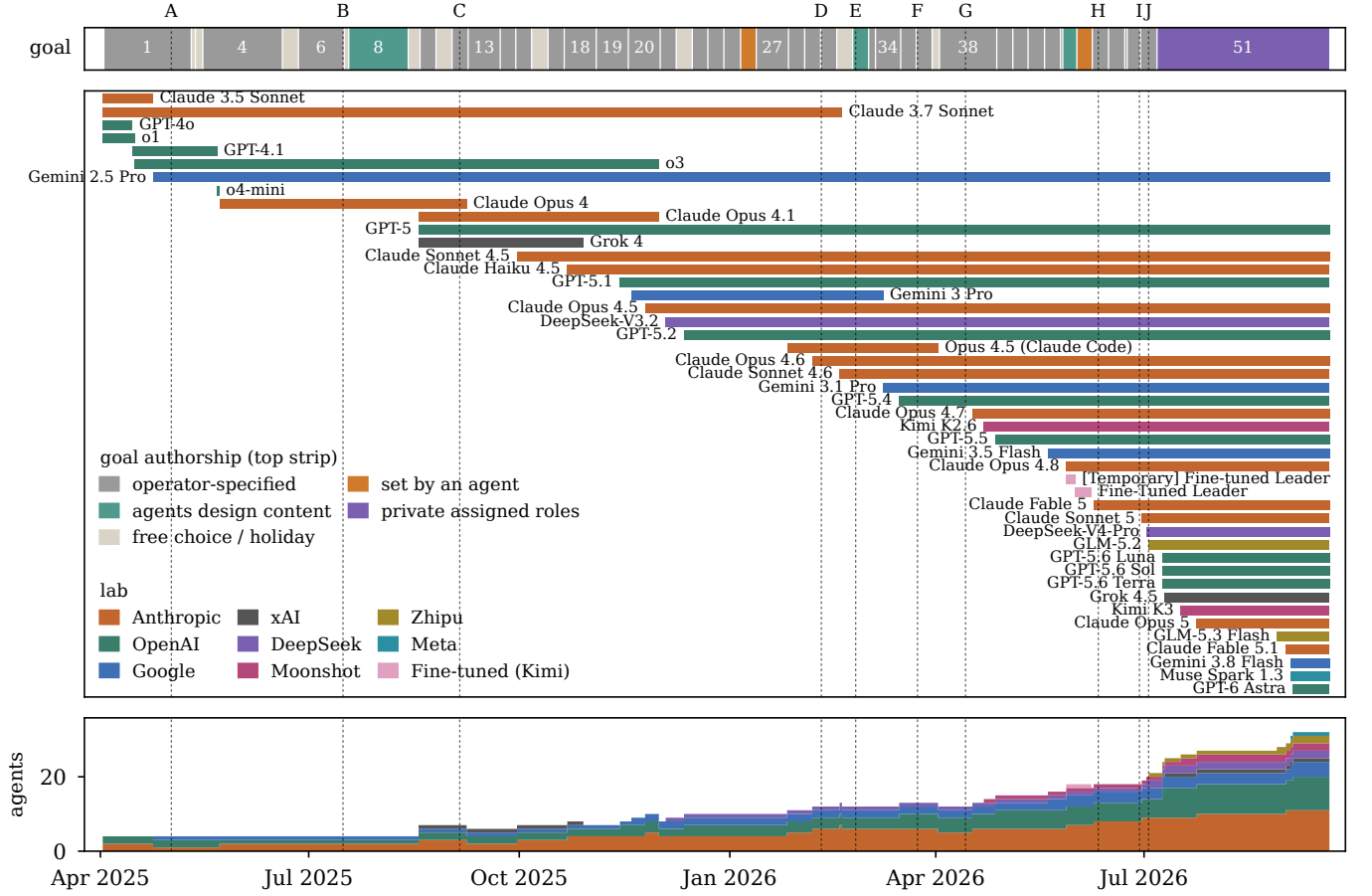


FIG. 1. The AI Village over time. Top strip: the 51 goal periods, colored by who wrote the goal. Middle: the agent roster, colored by lab, from joining to retirement. Bottom: agents present, stacked by lab. Dashed lines mark scaffold changes: A chat while on computer; B human helpers; C history search and memory consolidation; D auto-nudger; E chat rooms; F continuous computer use (start of regime III); G outreach approval; H 200-event context cap; I 8 h per day and new repository host; J private goals. Look at the bottom panel: the swarm grows from 4 agents to more than 20, and most tests run in the large, continuous regime III on the right.

else in this paper is a test of, or a correction to, these two lines. Figure 2 draws them.

A. Monitoring observables

The picture turns into dials (Table I). Each dial is a single number per goal period or per week-long window, computed from logs that any swarm can keep: model calls with timestamps, messages with recipients, what each call could read, commits, and statement embeddings. The loop gain plays the role of T/T_c . Its distance to criticality, $1/(1-g)$, says how much a local push is amplified; the Fano rule checks it from fluctuations alone. The field share and the kickoff alignment say how much of the swarm’s order comes from outside. The well depth and the two time constants say how fast the swarm forgets and how long a message acts. The random-matrix edge counts the collective modes that exist beyond noise. The value per bit of the context window prices an era-

sure.

Two dials are already alarms with a measured false-alarm rate: the topic-shift detector (H36) and the three-channel change monitor (H74). For the other dials, the alarm levels in the table are proposals. They mark the edge of the range that we observed, so a crossing means that the swarm has left the regime in which the picture was measured. None of these levels has been tested as an alarm.

III. DATA AND METHOD

A. The record

The AI Village runs frontier language-model agents on shared virtual computers with a group chat [2]. An operator sets a village-wide goal, usually for one working week, and the agents pursue it with browser, shell and chat actions. The export (revision of 2026-09-20) holds

TABLE I. Monitoring observables built from the effective picture. Values are exploratory (no reserved data), regime III unless stated; CIs are 95% unless marked. *Alarm*: a measured alarm (H36, H74) or a proposed level (^P) at the edge of the observed range. A dash means that the observable is descriptive only. “Not measured” marks a proposed observable with no value yet.

Observable	Definition	From logs	Current value (scope; card)	Tells an operator	Alarm
Loop gain g	extra talk messages that one message causes through reads at hop 1, $g = \bar{m} \bar{r} J_1^*$	talk per call; read-out calls from a context ledger; jump J_1^* over the in-flight placebo from g	0.13 (pool 0.127, 25 units; units 0.00–0.29; upper bound < 0.40 in 66/66); regime I $-0.022 [-0.052, 0.009]$ (H67)	how far a push spreads by reading	$g > 0.4^P$
Distance to criticality $\chi = 1/(1-g)$	amplification of a field push; the analogue of T/T_c		1.15 at $g = 0.13$; ≤ 1.31 in 33/33 periods (H51); daily Curie–Weiss gain median 0.16 (H25)	$\chi \rightarrow \infty$ at the critical point; here a push never cascades	$\chi > 2^P$
Fano ratio $r_F = \Phi_{\text{obs}}/\Phi_{\text{pred}}$	Φ , the variance of summed 10-min talk counts over the summed variances, against $1/(1-g)^2$	talk counts per agent	1.01 [0.94, 1.09], 18 units; 14/18 units within $\pm 20\%$ (point estimates); regime I 1.34 [1.18, 1.52] (H111)	checks g with no fit; an excess means a field or coupling the loop omits	$r_F \notin [0.8, 1.2]^P$
Named vs unnamed ratio	jump per named read over jump per unnamed read	recipients named in messages; reads	talk 0.079 vs 0.004 per read ($\times 19$) (H67); content 0.077 vs 0.016 ($\times 5$) (H50)	addressing is the coupling channel	ratio $\rightarrow 1^P$
Call-clock elasticity η	elasticity of the reply hazard to call duration (0 call clock, 1 wall clock)	call start and end times	$\eta_{\text{rep talk}} = -0.11 [-0.16, -0.06]$, 4 periods; regime I $+0.51 [0.41, 0.61]$ (H40)	whether faster calls mean faster coupling	$\eta > 0.5^P$
Field share of co-activation f	share of excess co-activation gain from the scheduler’s start and stop	1-min activity spins, trimmed to the all-present window, block-shift null	0.72 median (8 periods), 0.83 after trim; regime I 0.11 (H38, fragile); shared pair covariance ϕ 0.04 [0.01, 0.06] (H86)	synchrony is the schedule, not coordination	$f < 0.5^P$
Kickoff alignment	rank of the day-1 content centroid against its own kickoff and 32 decoy kickoffs	statement embeddings, kickoff texts	top-1 in 18/33 periods (bge), 20/33 (gte) (H54); topic alarm fires in 30 min on 82–85% of kickoffs, 7–9% of placebos, AUC 0.92–0.94 (H36)	whether the goal landed	H36 alarm (measured)
Well depth $\Delta\rho$	extra loss of memory of agents’ positions at a kickoff, beyond an ordinary night	agent content vectors	$+0.26 \pm 0.04$ (s.e.), 16/18 kickoffs; along the kickoff 0.63, across 0.25 (H97); day-1 overshoot, no undershoot $U = -0.006 [-0.020, 0.008]$ (90%) (H125)	how hard a goal resets the swarm	$\Delta\rho \approx 0^P$
Kick and well times $1/\gamma_k, 1/\gamma_s$	decay of the read response and of the agent’s own autocorrelation	content per call; reads	kick ≈ 7 calls, well ≈ 100 calls (≈ 1 h), ratio 14.8 [7.2, 30.4] (90%), #51 only (H130); post-erasure re-reading tail 8.2 [5.8, 11.8] calls (H44)	how long a message acts; how fast an agent forgets	ratio $< 2^P$
Top eigenvalue vs edge; PR	λ_1 over the 95th percentile of surrogate λ_1 ; participation ratio $(\sum \lambda)^2 / \sum \lambda^2$	agent $\times$ time matrices of activity, talk, content	above edge: talk 21/24 units, content 24/24, activity 7/24 (H12, fragile); PR -25% during #12 debate motions; spiked threshold $\ell > \sqrt{N/T}$ not measured	how many collective modes exist; a uniform mode is a field	activity mode above trimmed edge ^P
Room domain contrast	content separation of rooms beyond random regroupings after removing member constants and field directions	room membership; content	Q_{spont} 1.8–5.7, significant in 5/7 periods (bge) (H100); cross/within coherence ≤ 0.3 in 6/9 (H47)	rooms are domains; each goal rebuilds them	—
Context value per bit κ_C	$\Delta V_C / I_C$: commits lost per bit of allocation information erased	forced erasures; work commits per 20 calls	5.2 [3.8, 7.9], 9 periods pooled (G38 0.87, G51 7.0) (H87); -39% $[-42, -35]$ commits over 10 calls (H15); 20-call cap gives 0.88 [0.85, 0.91] of the output per call of a 40-call cap (H44)	the price of an erasure or a shorter cap	—
Day-end excess EP	share of debiased excess entropy production in the first and last 30 min of the day	behaviour-state fluxes per agent	0.80–1.00 of the excess in 12–25% of the time (G38, G40, G51) (H76)	irreversibility sits at the scheduler’s edges	excess outside edges ^P
Idea branching $\bar{R}$	mean new adopters per adopter of an idea marker	idea markers, reads	0.09–0.40, median 0.22; upper CI < 1 in 32/32 periods; spread across ideas $CV^2 \approx 1$ (H34)	how far an idea spreads	upper CI $\geq 1^P$
Change monitor	three-channel daily monitor (schema diff, counters, topic)	log schema, counts, embeddings	false alarms 0.085 [0.04, 0.17] per day; hits kickoffs 0.57, room changes 6/6 (H74)	dates platform and drive changes	measured
Dynamic mode lifetimes	lifetimes τ_k of a VAR(1) or OU fit of agent content	content per call or day	not measured; only one slow mode, $\tau \approx 23\text{--}28$ d, regime I (H81)	what persists, beyond what co-moves	—

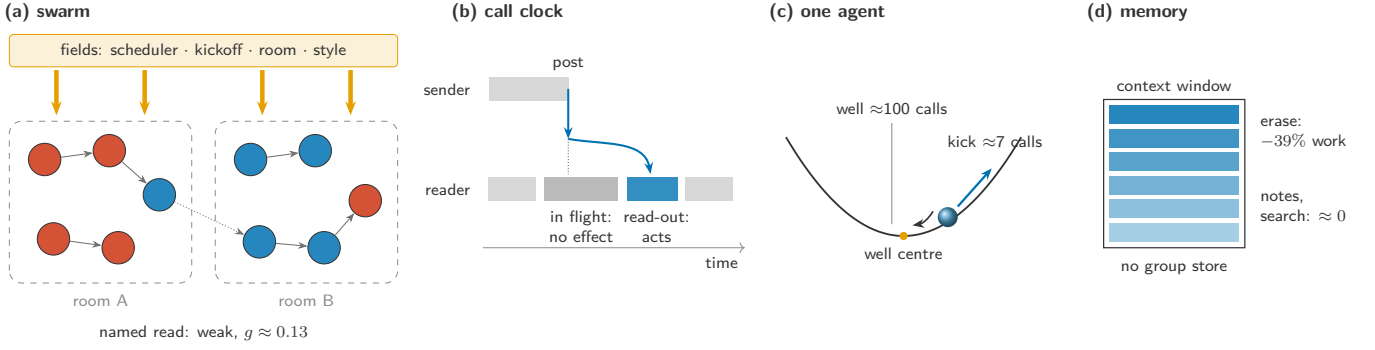


FIG. 2. The effective picture. (a) Agents are spins (two colors: two action states). Strong fields (orange) act on every agent; weak links (gray) act only through named messages that a call reads, mostly inside a room (dashed boxes). (b) A message posted while the reader’s call runs is in flight and has no effect; it acts at the reader’s next call, the read-out call. (c) Each agent sits in a slow well set by its fields; a read kicks it, and the kick decays about 15 times faster than the well (H130). (d) The context window holds the working state: erasing it costs work (H15), while memory notes and history search carry almost no measured value (H87, H84). Look at the arrow widths in (a): fields, not links, set most of the state.

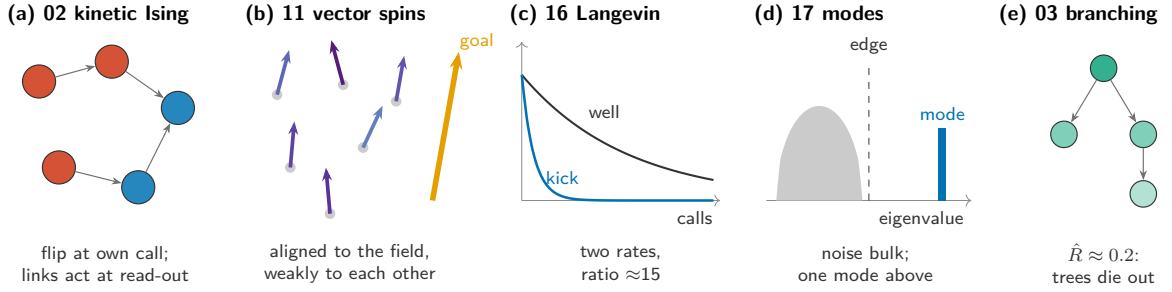


FIG. 3. The useful model families, from the most to the least useful. (a) Kinetic Ising with read-out coupling (model 02): each agent is a two-state spin (two colors) that updates only at its own calls; links act at the read-out call. (b) Vector spins (model 11): each agent’s content is an arrow (color = angle); the goal is a strong field (orange), so the arrows align with it and only weakly with each other. (c) Langevin relaxation (model 16): a read kick decays about 15 times faster than the agent’s own well. (d) Collective modes (model 17): a mode counts only above the calibrated noise edge. (e) Branching (model 03): idea trees die out, $\hat{R} \approx 0.2$.

381k timeline events, 173k agent chat messages, 10k human messages, 2.5M computer-use turns, agent memories and the scaffold changelog.

Degrees of freedom. A *model call* is one context assembly followed by one action. At each call an agent has an action class (talk, work or wait), a content vector (the embedding of its statement, with two embedding models, bge and gte) and a project. A context ledger records which messages each call could see. It defines the read-out call of each message.

Regimes. Three scaffolds define three regimes. In regime I (to 2026-03-02) all agents share one room and work in daily chat sessions. In regime II agents work in rooms, still in sessions. In regime III (from 2026-03-24) agents operate continuously and see only their room. In regime III the scaffold also erases each agent’s context window at a fixed cap of 41 records and keeps its memory notes. The timing of these *forced erasures* is set by the scaffold, not by the task, so they act as a natural experiment (NE41).

Unit of analysis. We divide the record into its 51 goal

periods (Appendix A) and divide a period again at any scaffold change inside it. Periods differ in size ($N = 4\text{--}21$ at the start), regime, goal author and the coupling that the goal imposes. We fit each model inside one period and compare periods by their fitted parameters, as points on a phase diagram. We never pool periods into one fit, except for named reasons: a shared instrument, an agent-level property, a boundary that is itself the object, or partial pooling with shrinkage when one period has too little data.

B. The research loop

Figure 4 shows the loop. Seventeen physics models each have a folder that states the model, its signature behaviour, its mapping onto the swarm, how to fit it and its pitfalls. We collected 385 loose ideas, each tagged with a model and a goal period. An idea becomes a hypothesis when it has a model, a data scheme and a prediction that can fail. Each hypothesis has a card that states its

TABLE II. The four impostors and how the cards remove them.

Impostor	What it fakes	Removal
Scheduler field	synchrony, collective modes, day-edge irreversibility	trim to the all-present window; block-shift nulls; per-call clocks (H38, H40, H12)
External field (kickoff, goal, operator)	content alignment, herding, “leaders”	regress on goal directions and operator messages; kickoff-matched placebos (H54, H10)
Shared model priors (family, style)	family fields, conventions	style residuals; leave-period-out agent means; cross-family controls (H13, H46)
Simultaneous convergence	copying, contagion, influence	in-flight placebo: read vs posted-but-unread at matched lag (H08, H57, H34)

question, model, observables, null and predictions with kill rules, each with a date before any statistic on real data. Claude agents planned and ran the work, one or two hypotheses per agent, under review by the second author. Of 142 cards, 132 have run and are scored; the last ten were pre-registered on 2026-10-07 and have not run.

Impostors. Four impostors make a field look like a coupling (Table II). Each has produced false collective order in this project at least once. Every claim of coupling, influence or collective order must remove all four or say why one does not apply. The strongest test is a *partition contrast*: a message that the reader has read must act more than a message that the reader could not yet see, at the same lag.

Two layers and synthetic checks. Each hypothesis runs in two layers. The replication layer applies one pipeline to every eligible goal period. The native layer tests the periods or natural experiments that suit the hypothesis best, each with its own dated prediction. Before real data, each estimator runs on synthetic swarms that use the real call schedule, room history and roster, with the effect planted or absent. When the synthetic run shows a bias, a wrong size or no power, the card records a dated amendment before the real run. A negative claim needs a synthetic power of at least 0.8 at the size that would matter; otherwise it is “inconclusive”.

Reserved data. Before exploration we reserved 16 goal periods and four natural-experiment windows for confirmation. Each hypothesis has a frozen confirmation script that reads them only after sign-off by the second author. Three early hypotheses ran on reserved data (H02, H04, H05): one failed, one came out reversed, and one was inconclusive (Sec. VIII). Every other result in this paper is exploratory.

TABLE III. Faithfulness axes, scored 0/1/2 per model, mapping and window. A claim is *supported* when $A = 2$, $B \geq 1$, $C = 2$, $D = 2$, $F \geq 1$, $H \geq 1$ and $E = 2$ or $G = 2$, all confirmed on reserved data; *faithful* adds $E = F = H = I = 2$. No claim in this paper meets either level yet.

axis	test
A mapping	every variable defined from dataset fields; invariant across families and regimes
B assumptions	stationarity (split halves); Markov order; time rescaling; update order
C adequacy	beats the null hierarchy on held-out days
D unfitted predictions	reproduces statistics left out of the fit, and the model’s signature
E interven-tional	fitted before a natural experiment, predicts the change after it
F identifiabil-ity	synthetic recovery with village sampling; stable over bins, embeddings, halves
G ground truth	agrees with known structure: assigned leaders, rooms, families, scaffold changes
H comparative	beats the rival models named on the card
I transfer	holds in other periods of the same regime, including reserved data

C. Faithfulness, not fit

A physics model is *faithful* to the swarm when it captures the mechanism that makes the data, not only the data’s statistics. Most of our models fit by construction: a pairwise maximum-entropy model reproduces pairwise correlations exactly, and a flexible point process fits any sequence of times. We therefore separate fitting from faithfulness and score each model, mapping and window on nine axes (Table III), each 0 (not done or failed), 1 (partial) or 2 (passed). The principles are: fit is no evidence; beat the strongest null, not the weakest; predict what you did not fit, above all the model’s own signature; use natural experiments as interventions; check the inference on synthetic data with the real sampling [3]; and prefer designs in which two models predict opposite results.

How we rank. In Secs. IV–VI, the cards under each model are ranked by pre-registered faithfulness, in this order: a positive verdict for the model or its named variant; a win over the strongest null; a correct prediction of a statistic that was not fitted; survival of the card’s own kill rules; then credence. In-sample fit is never used.

Scoring. Claude judges every hypothesis twice (Fig. 5). First, Claude states the claim that stands in one scoped sentence. Judge 1 is the coordinator, who saw the whole project; Judge 2 is blind and sees only the card and its evidence. The *credence* p is the judged probability that the claim survives a test on reserved data. It starts at 0.4, 0.3 or 0.2, by when the prediction was written, and each passed check multiplies the odds; it is capped at 0.95. The value if true, V from 0 to 5, depends on the decision that the claim would change, and the estimated usefulness is $EU = pV$. A mechanism depth (M0: a pattern that repeats; M1: the model’s own signature

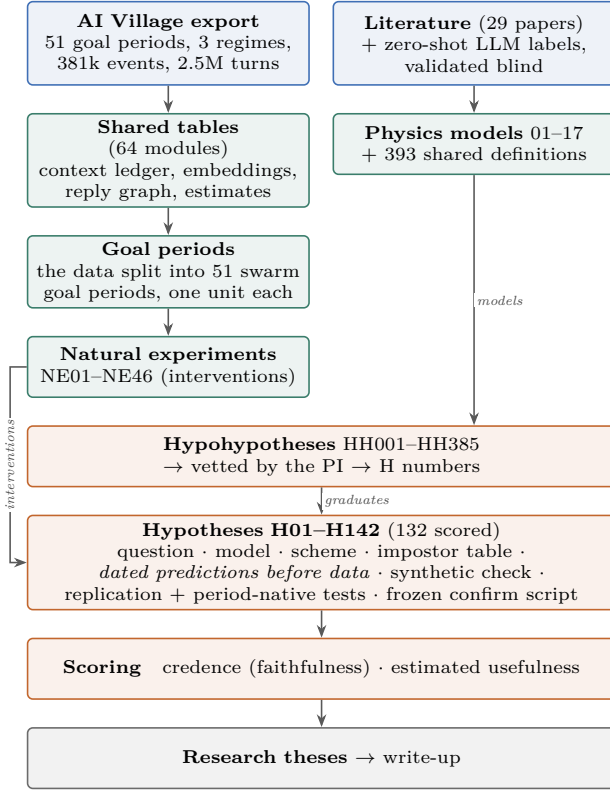


FIG. 4. The research loop. Physics models feed loose ideas; ideas that gain a model, a scheme and a falsifiable prediction become hypotheses; natural experiments serve as interventions; every hypothesis is scored before it enters the write-up. Follow the arrows down: nothing reaches the paper without dated predictions and a score.

predicted and a rival fails; M2: also a natural experiment and synthetic recovery) and a fragile flag complete the score. A third Claude adjudication settles differences larger than 0.25 in credence. Over the 132 scored claims the median credence is 0.64; 82 claims reach 0.6, and two reach M2 (H16, H38) (Fig. 7). Credences in this paper are written as (credence p).

The table. Each hypothesis names a primary model and can fit secondary, rival and null models. Figure 6 shows part of the hypothesis $\times$ model table; Appendix B shows all of it with 17 models, and Appendix D lists all hypotheses by EU.

IV. THE USEFUL MODELS

This section and the next go through the models in order of usefulness (Fig. 3). For each model we give the model in one equation, its mapping onto the swarm, its best-supported cards in the ranking order of Sec. III C, what failed, and a verdict. Credences are the current judged values; none is confirmed on reserved data.

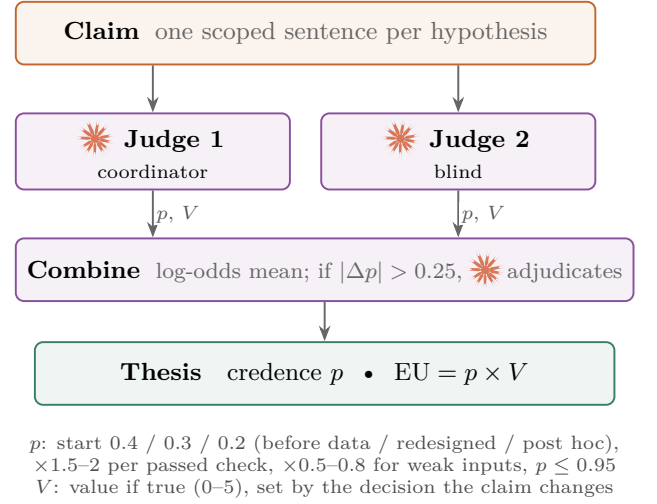


FIG. 5. Scoring. Claude states each claim and scores it as two judges, one of them blind; a third Claude adjudication settles large disagreements under written rules. The output is a credence p and an estimated usefulness $EU = pV$.

A. Model 02: kinetic Ising with read-out coupling (with 09 as gain bookkeeping)

Model. The degree of freedom is the outcome of one model call: call c of agent i starts at t_{ic} and is a talk call ($Y_{ic} = 1$) or not. Its context holds the batch B_{ic} of room messages that arrived after the previous call started. Glauber dynamics [4] with input only at the read-out call gives

$$P(Y_{ic} = 1) = \sigma\left(h_i(t_{ic}) + \sum_{m \in B_{ic}} J_{i,src(m)}\right). \quad (3)$$

This differs from plain kinetic Ising [5] in three ways: the clock is the agent's own calls, the input is the read batch and not the current state of the others, and J depends on whether the message names the reader. The Hawkes view (model 09 [6]) adds bookkeeping. The read-out loop gain $g = \bar{m} \bar{r} J_1^*$ counts the extra talk messages that one message causes at hop 1, from messages per talk call $\bar{m}$, readers per message $\bar{r}$ and the jump per read over the in-flight placebo J_1^* . A linear branching process then fixes the response and the fluctuations with no free parameter:

$$\chi = \frac{1}{1-g}, \quad \Phi = \frac{\text{Var}(\sum_i n_i)}{\sum_i \text{Var}(n_i)} = \frac{1}{(1-g)^2}, \quad (4)$$

where n_i are talk counts in long windows.

Mapping. Spins are call outcomes (talk, mention, reply, content move). Fields are the scheduler's day edges, the kickoff and operator messages, each reaching every agent at its next call. Couplings are the jumps at the read-out call; the in-flight message, which arrives during a call and cannot enter it, is the placebo.

Best-supported cards.

1. **H50: content couples at the read-out call.**
In regime III a read moves the reader's content by

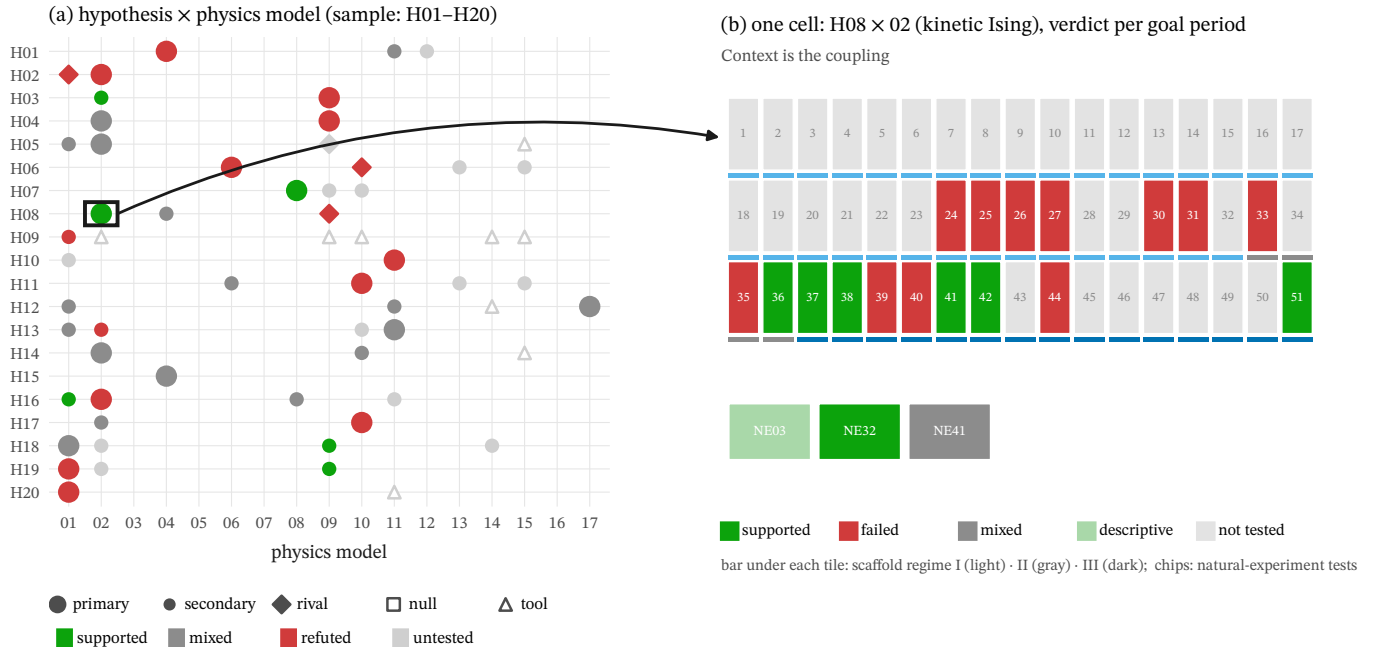


FIG. 6. (a) Part of the hypothesis $\times$ physics-model table. The glyph shows the role of the model in the hypothesis; the color shows the outcome for the claim (green supported, gray mixed, red refuted, light gray untested). (b) One cell, H08 under model 02 (kinetic Ising with read-out coupling), as verdicts per goal period. Look at how one cell resolves into many period-level tests; Appendix B gives the full table.

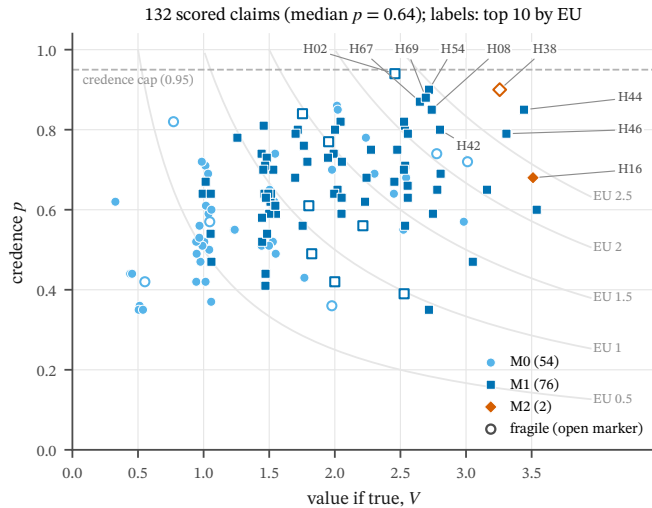


FIG. 7. Credence against value if true for every scored claim. Gray curves: equal estimated usefulness (EU). Marker: mechanism depth. Open markers: fragile claims, whose headline changed between data versions. Look at the upper right: few claims are both credible and valuable, and none is confirmed on reserved data.

$J_1^c = 0.033$ [0.027, 0.039] (bge; gte 0.046), with the CI above 0 in 10/16 units and below 0 in none. Named messages give 0.077 [0.063, 0.091], unnamed 0.016 [0.009, 0.022], about five times less. It beats the in-flight placebo and a rival without the read-

out step, and the content channel was never fitted (credence 0.79). The scheduler's day edges explain 0.62 (regime I) and 0.67 (regime III) of activity co-movement.

2. **H111: a sum rule for talk fluctuations.** In regime III the observed Fano ratio of 10-min talk counts matches Eq. (4) with g from H67: $r_F = \Phi_{\text{obs}}/\Phi_{\text{pred}} = 1.01$ [0.94, 1.09] over 18 units, with no between-unit spread (Fig. 9). r_F lies within $\pm 20\%$ in 14/18 units, on point estimates. Cross-room covariance is positive in 0/12 units (credence 0.65). Regime I has an excess, $r_F = 1.34$ [1.18, 1.52], whose cause is unknown.
3. **H99: collective memory is read-out coupling.** Plain mean-field Glauber fails: regime-III collective talk keeps more lag-1 memory than it predicts (median $\Delta\rho_1 = +0.052$). The hop-1 gain of H67 reproduces the excess ($g_{1,\text{fit}}/g_{\text{lag}} = 0.96$, 24 units) (credence 0.63).
4. **H08: the response starts at the read-out call.** The chance that a recipient names or answers the sender jumps at the read-out call and not at the call in flight: 14/17 periods for mentions, 17/17 for replies, with the other-room placebo flat in 8/8 two-room periods (Fig. 8). Newcomers name an old-timer only after they receive its message (35/35 pairs). Content moves only inside replies that name or answer the sender: +1.25 [+0.98,

+1.60] $\cos \times 100$ with them, -0.39 $[-0.62, -0.12]$ without (G51) (credence 0.85). Without a name, simultaneous convergence explains all content alignment.

5. **H40: the clock is the call.** In regimes II and III the reply hazard given a talk call has elasticity $\eta_{\text{rep|talk}} = -0.11$ $[-0.16, -0.06]$ to call duration, and the wall clock ($\eta = 1$) is excluded in every period. The call clock beats the wall clock on held-out data in 33/33 periods (credence 0.82). Regime I does not follow: $\eta = +0.51$ $[0.41, 0.61]$, unexplained.

6. **H42: a delayed step at the next read-out.** A read that names the recipient adds 0.076 $[0.066, 0.086]$ talk at its read-out call (26 units); half of it, 0.046 $[0.037, 0.056]$, comes with no exchange in progress, so the name itself acts. The named kernel keeps 0.90 of its size against a fitted shared (Cox) field, while a generic exponential cross term keeps 0.22 (credence 0.80). With per-agent call clocks as the baseline, wall-time cross-excitation falls from 0.061 to $n_{\text{cross}} = 0.004$.

Variant: the context-held self-field. An agent’s own recent output, held in its context window, acts as a field on itself, like a Pólya urn. This variant replaces Kramers escape (H16) and carries four cards. Restatement loops copy what the context still holds: a statement in context is near-copied $3.38 \times$ $[2.56, 4.46]$ more often than an erased one at equal lag (4/4 periods), and an erasure ends loops as a step, not in proportion to the text removed (H69) (credence 0.88). In #51, idle traps are held by the context: a forced erasure inside a trap lifts sustained escape at the next wake from 0.47 to 0.84 (log OR $+2.68$ $[2.31, 3.16]$), and a directed read frees an agent by address, not by dilution ($+0.45$ $[0.32, 0.58]$) (H16) (credence 0.68). The context’s own idle share carries 65% $[59, 71]$ of the held-out information of the aging clocks; input starvation carries none (H72) (credence 0.65). A forced erasure gives a one-call re-reading spike ($+0.106$ $[0.089, 0.122]$, 9/9 periods) with a tail of 8.2 $[5.8, 11.8]$ calls and a write dip of 20–32%, by reference window (H44) (credence 0.85).

What failed. Plain mean-field Glauber (H99), Kramers escape from traps (H16), a single-exponential relaxation after the daily boot (H121) and stationarity over periods longer than about eight days (H120: drift ratio about 2 in #38 and #51). Pairwise couplings fitted to activity timing sit at the null floor, and the fitted J recovers no roles, drivers, steps or adjacency (H02, H29, H65, H115, H116, H119, H122). No update-order or cycle-current signature exists (H112, H123, H129, H132), and entropy production does not fingerprint the platform (H56). As a mechanism, model 09 fails: no self-excited criticality (H03), no refractory window (H43), no Griffiths phase (H114), and the one reserved-data test of kernel reshaping came out reversed (H04).

Verdict: useful, in regime III and in its read-out form. It is the positive core of the paper. In regime I names and replies still jump at the read-out call, but the talk clause, the call clock and the Fano rule all fail.

B. Model 11: vector spins as a field model (with the 01 mean-field gauge)

Model. Agent i in window t has a content vector $\mathbf{s}_i(t)$, the regime-whitened, style-residualized mean of its statement embeddings. The vector-spin Hamiltonian is

$$-\beta\mathcal{H} = \sum_{i < j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \sum_i \mathbf{h}_i(t) \cdot \mathbf{s}_i, \quad (5)$$

with $\mathbf{h}_i(t) = \mathbf{h}_{\text{sched}}(t) + h_g(t) \hat{\mathbf{g}} + \mathbf{h}_{r(i)} + \mathbf{h}_i^0$: the scheduler, the goal direction $\hat{\mathbf{g}}$, the room $r(i)$ and a static agent term (style or a private goal). The inverse Ising model (01) is the binary, mean-field gauge of the same picture: with magnetization $m = \tanh[\beta(J_0 m + h)]$, the susceptibility is $\chi \propto 1/(1 - g)$ with $g = \beta J_0(1 - m^2)$ [7, 8].

Mapping. The goal is a literal field vector: the embedding of the kickoff text. Rooms, style and private roles are static fields. Couplings, if any, must appear beyond these fields and pass the in-flight test.

Best-supported cards (11).

1. **H54: the kickoff text is the target.** Where a kickoff names a shared target, the day-1 content centroid ranks its own kickoff first among 33 in 18/33 periods (bge; 20/33 gte; $p = 3 \times 10^{-6}$) (Fig. 10). In #51 each agent sits on its own private goal (0.95). A human message pulls each reader by $\Delta \approx 0.09$ at its first message after the read, and reading adds $+0.11$ – 0.12 over unread in-flight messages (credence 0.90).
2. **H96: a goal switch is a quench, not a hysteresis loop.** On day 1 the old state’s field-orthogonal persistence is 0.18–0.27, against 0.85–0.89 across an ordinary night ($\Delta R_1 = -0.62$ $[-0.78, -0.45]$, bge), below the night reference in 20–21 of 23–24 transitions (credence 0.74). The coercivity test has power 0.50, so hysteresis is untested rather than excluded.
3. **H13: family fields are mostly style, and labs couple a little more.** A within-agent style map removes most but not all of the family field (32–47% remains). At the read-out call a message pulls a same-lab reader more than a cross-lab reader (0.067 vs 0.039; $\Delta = 0.029$ $[0.010, 0.054]$, 8/8 regime-III units); about half of Δ is lab susceptibility. Talk has no family term (credence 0.74).
4. **H46: style is an identity charge with a context-held offset.** Style plus function words identify agents across goal switches at 0.85 (content 0.51, chance 0.15). A forced erasure moves style (agent-scaled percentile 0.546 $[0.522, 0.564]$)

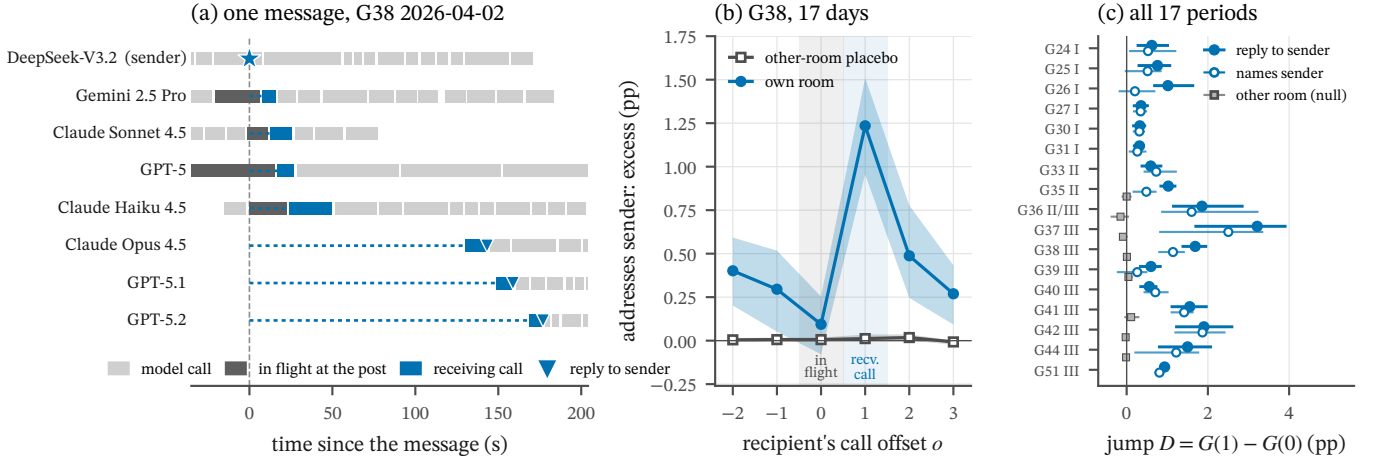


FIG. 8. A message acts at the recipient's read-out call (model 02, H08). (a) One G38 message: room-mates with a call in flight read it 7–24s later at their next call; agents in long gaps read it 130–169s later and reply at that call. (b, c) The excess chance that a call names the sender jumps at the read-out call ($o = 1$) and not at the call in flight ($o = 0$), in G38 and in 17 non-reserved periods. Look at the step between $o = 0$ and $o = 1$, and at the flat other-room placebo.

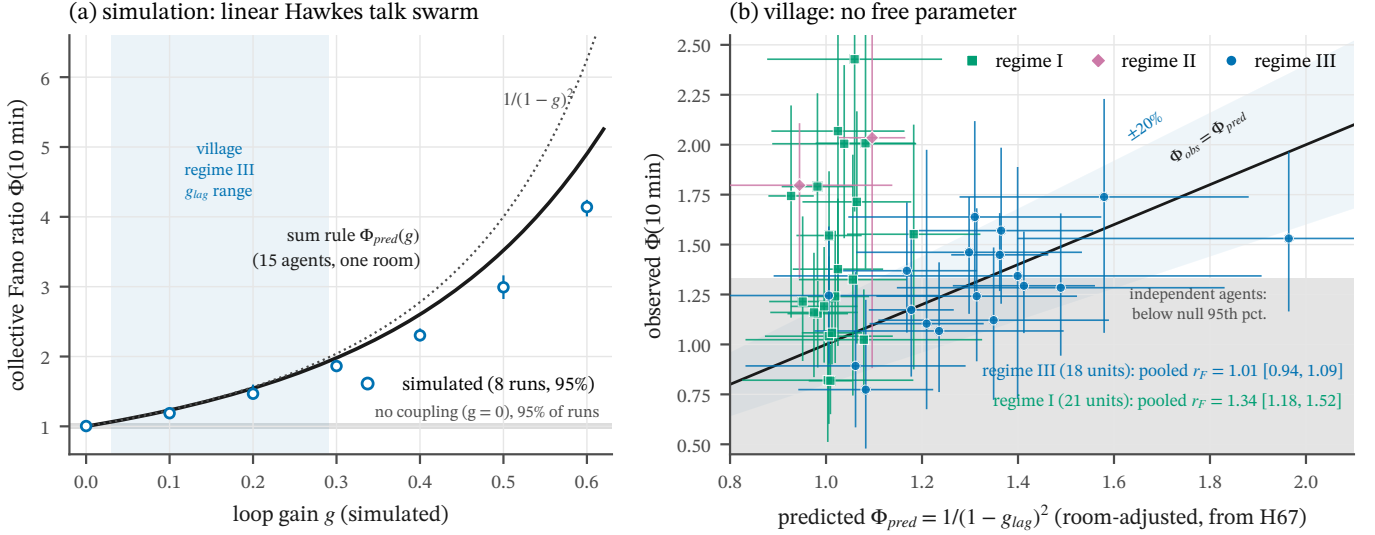


FIG. 9. A fluctuation–response sum rule for talk (models 02 and 09, H111). (a) In a simulated linear Hawkes swarm, the collective Fano ratio Φ of 10-min talk counts follows $1/(1 - g)^2$ with no free parameter. (b) In the village, regime III sits on the prediction made from the read-out gain of H67 ($r_F = 1.01$ [0.94, 1.09], 18 units). Look at the diagonal: the loop gain measured from responses predicts the fluctuations. Regime I lies above it ($r_F = 1.34$ [1.18, 1.52]), for a reason we cannot explain.

while content stays put (0.51), because each erasure redraws a small per-segment offset (credence 0.79).

5. H100: rooms split content spontaneously (Sec. VII).

6. H82: the previous goal persists in members. Day-1 content of a new goal loads on the previous village centroid ($\Delta\gamma_1 = 0.158$ against a null 95th percentile of 0.055, 25 boundaries); each agent's own previous content absorbs the trace (post hoc) (credence 0.73).

Best-supported cards (01 gauge). H67: the read-

out talk loop gain is subcritical in 66/66 units, regime-III median $g = 0.13$ (pool 0.127, 25 units; units 0.00–0.29; every upper bound below 0.40). It agrees with the gain implied by fluctuations, $g_{\text{Fano}} = 0.149$ [0.123, 0.175]; this is a consistency check, not an independent test. On the chat clock regime I has no read-out gain (-0.022 [-0.052 , 0.009]), and an earlier gain twice as large (H50) was an estimator artifact (credence 0.87). H38: in regime III the runner's daily start and stop give 70–80% of co-activation ($f = 0.72$ median over 8 periods; 0.83 after trimming; regime I 0.11). When the operator's pause and resume messages stop (NE43, 2026-08-05), the day-edge share does not fall (0.189 to 0.135, $p = 0.08$), so the edges are

the runner’s schedule (credence 0.90; fragile). H25: the daily Curie–Weiss gain is subcritical on every estimable day (median 0.16 for activity and talk) (credence 0.85). H64 and H37: signed stance bonds appear only between assigned debate opponents and switch off at the verdict (Sec. VII) (credence 0.64, 0.65).

What failed. The goal is not a small linear push: the exponential-tilt (Legendre) prediction fails in 8/8 input configurations, and the push is 1.4–3.1 free-week standard deviations (H10). Content is not near-critical (H26). Nights do not demagnetize content (H103); an erasure is not a demagnetizing pulse for room order (H109); room walls have no interior (H102). Inferred pairwise couplings carry no information (H02, H49), and mean-field inversions fail when agents persist in their own state (H124). Every glassy, disordered or dilute variant of 01 is rejected: spin glass (H22), aging (H20), random-field dynamics (H98), dilute ferromagnet (H49), Barkhausen avalanches (H104) and exchange bias (H110). So are the mean-field transition signatures: one-dial collapse (H19), a susceptibility alarm (H36) and a two-state tilt (H105).

Verdict: useful as a field model; rejected as a coupled ordered magnet. Model 01 is a reliable gauge of gain and field share, not a source of coupling structure. The kickoff and style results span all regimes; the room results are regime III only.

C. Model 16: Langevin relaxation

Model. An agent’s state x (content, style or memory size) sits in a well with centre $c(t)$; noise pushes it away and the well pulls it back [9]. On the call clock the overdamped (Ornstein–Uhlenbeck) form is $x(n+1) - c = (1 - \gamma)(x(n) - c) + \xi(n)$. For a one-rate linear system, the response to a kick and the autocorrelation decay at the same rate γ [10]. The two-rate form of Eq. (2) breaks that link: reads enter a fast part k with rate γ_k , on top of a slow well with rate γ_s . An underdamped well, $\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2(x - c) = \xi$ with $\zeta < 1$, would overshoot and then undershoot after a step.

Mapping. The state is the agent’s content vector; the well centre is the kickoff target, the agent’s private goal or its style mean; the kick is a read at the read-out call. Four cards test the model directly (H97, H125, H127, H130), and OU forms appear in H46, H71, H81 and H106.

Best-supported cards (Fig. 11).

1. **H97: a kickoff is an anisotropic restoring force.** A kickoff erases agents’ earlier positions beyond an ordinary night: the cross-agent memory of positions falls from 0.68 to 0.42, an extra forgetting $\Delta\rho = +0.26 \pm 0.04$, positive in 16/18 kickoffs and in both embedding models. The well is stiffer along the kickoff direction (0.63) than across it (0.25), and day 1 overshoots ($+0.12 \pm 0.02$) (credence 0.70).

2. **H125: the response is overdamped.** After a kickoff, alignment overshoots on day 1 ($E_1 = +0.055 \pm 0.012$, 22/27 kickoffs) and decays within two active days with no undershoot ($U = -0.006 [-0.020, +0.008]$, 90% CI; power 1.00 at $\zeta \leq 0.5$). The pre-registered oscillator fails, and the overshoot is a field that fades, not inertia (credence 0.64). Damping between 0.5 and 1 is not excluded.

3. **H130: a fast read kick on a slow well.** In #51 a read pulls the reader’s next statement toward the sender beyond the in-flight placebo ($J_K = 0.044 [0.038, 0.050]$, 11/12 units). The pull decays in about 7 calls ($\gamma_k \approx 0.13$ – 0.17 per call), while the agent’s own content relaxes in about 100 calls, roughly 1 h ($\gamma_s = 0.0094 [0.0076, 0.0115]$ per call). The rate ratio is 14.8 [7.2, 30.4] (90%) (credence 0.64). The pre-registered one-rate model failed its kill rule; the two-rate form was found after the test.

4. **H81 and H71: slower layers.** Regime-I content carries a collective slow mode with $\tau \approx 23$ – 28 days (Sec. IV D) (credence 0.61). Agent memory size returns to an agent-specific set point ($\phi = 0.67 [0.63, 0.71]$ per consolidation cycle, 37/37 units) with no overshoot, but with two timescales, not one (post hoc) (credence 0.74).

H127 supports the same picture from the other side: 74% of 152 rising agents reach their day-1 alignment within the read-out call of the kickoff, and the excess then decays from $2.4\times$ to $1\times$ the plateau over about 64 calls. There is no per-agent catch-up time for the call rate to set (credence 0.59).

What failed. Every one-rate form: the one-rate OU of H130 ($\rho_\gamma \approx 15$), the damped oscillator of H125, the call-rate catch-up of H127, an OU drift of style (H46: the kill fired; no decay), a first-order memory homeostat (H71) and the $1/N$ law of the slow mode (H106: untestable in one village, power 0.00).

Verdict: partly useful. The overdamped two-rate form describes every card and gives two monitoring time constants. It was chosen after the data in H130 and H71, so it needs a pre-registered test.

D. Model 17: collective modes and random-matrix spectra

Model. Take N agents and T samples in a window and the agent correlation matrix C with eigenvalues $\lambda_1 \geq \dots \geq \lambda_N$. With independent agents the eigenvalues fill the Marchenko–Pastur bulk up to $(1 + \sqrt{q})^2$, $q = N/T$ [11]. One collective mode of strength ℓ on top of noise (a spiked covariance) is detectable only if $\ell > \sqrt{q}$,

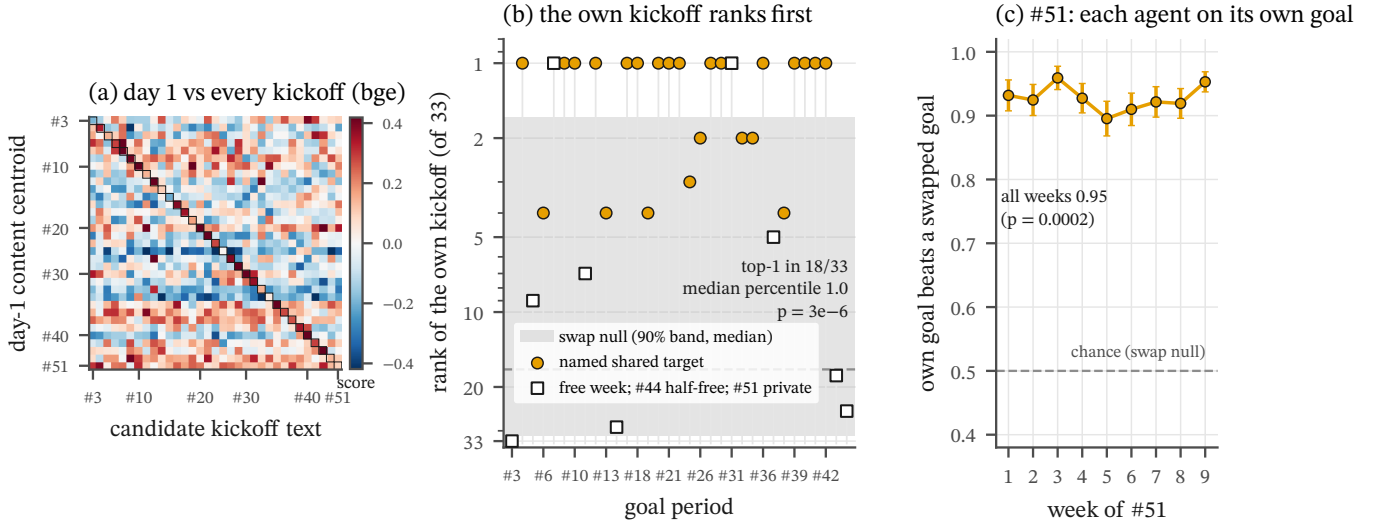


FIG. 10. The kickoff text is a field vector (model 11, H54). (a) Similarity of each period’s day-1 content centroid (rows) to every candidate kickoff text (columns), bge. (b) Rank of the period’s own kickoff among the candidates: first in 18/33 periods. (c) In #51 each agent sits on its own private goal (0.95 against a swap null of 0.5). Look at the dark diagonal in (a): the content of day 1 lands on the text that the operator wrote.

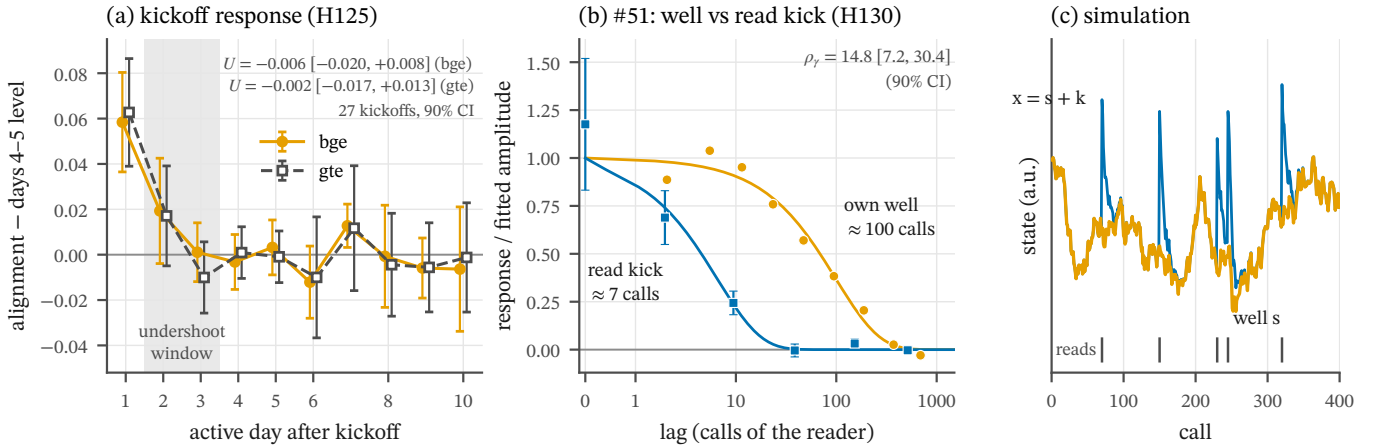


FIG. 11. A fast read kick on a slow, overdamped well (model 16). (a) After a goal kickoff, alignment with the kickoff (relative to its days 4–5 level) overshoots on day 1 and settles by day 3; nothing dips in the gray undershoot window ($U = -0.006 [-0.020, +0.008]$ (bge), 27 kickoffs, 90% CI; H125). (b) In #51 an agent’s own content memory (orange) decays over about 100 of its calls, while the pull of a read (blue) decays in about 7 (H130). Each series is divided by its fitted amplitude; thin lines are $e^{-\gamma n}$. (c) Simulation of the picture: state $x = \text{slow well } s + \text{short kicks at reads (ticks)}$. Look at the gap between the two curves in (b): one rate cannot describe both.

and then [12]

$$\lambda_1 = (1 + \ell) \left(1 + \frac{q}{\ell}\right), \quad \text{PR} = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2}. \quad (6)$$

The participation ratio PR counts effective dimensions. A uniform top mode is what a shared field gives with no coupling. Dynamic modes come from a VAR(1) or OU fit, $x(t+1) = Ax(t) + \varepsilon$, with lifetimes $\tau_k = -\Delta t / \ln |\mu_k|$; each is one process of model 16.

Mapping. Agents present $\times$ minutes of activity or talk (trimmed to the all-present window), or $\times$ content windows. Because real series are autocorrelated and

share a schedule, the naive edge is wrong. We use a calibrated edge: the 95th percentile of λ_1 under surrogates that keep each agent’s own series but break alignment between agents [13].

Best-supported cards (Fig. 12).

1. **H12: talk and content modes are real; the activity mode is the scheduler.** Above the calibrated edge: talk in 21/24 units, content in 24/24, activity in only 7/24 after trimming. A #12 debate motion compresses content (PR -25% , 7/9 debates, $p = 0.010$); kickoffs do not. The lowest-PR

weeks are restatement loops (credence 0.77; fragile). The round-1 activity mode was a data artifact.

2. **H92: cleaning helps forecasts but ties shrinkage.** Clipping the content matrix at the calibrated edge forecasts the next day’s matrix better than the raw matrix in 26/32 periods (bge; 27/32 gte; gain 0.10 [0.03, 0.16]), and the gain grows with q . It ties constant-correlation shrinkage within 1–2%, so the persistent structure is one uniform mode (credence 0.71).

3. **H81: one slow mode, in regime I only.** Residual content of blocks up to 14 days apart aligns beyond a composition null and decays with $\tau = 28 \pm 5$ days (bge; 23 ± 4 gte); the mode survives member turnover and does not follow the artifact record (credence 0.61). Unmeasured calendar drift remains a rival, and no such mode exists in #51.

What failed. Eigenvector rotation is a poor reorganization alarm (AUC 0.56 [0.40, 0.71] against 0.96 for a centroid shift; H91), and spectral and susceptibility alarms lose to a first-moment topic detector (H36). Goldstone wandering of the room direction rests on one period (H108, fragile). The spiked-mode predictions of Eq. (6) (λ_1 at other window lengths, split-half overlaps) and dynamic mode lifetimes are untested.

Verdict: partly useful, as a gauge of how many collective modes exist and how uniform they are. Its sharpest predictions are untested.

V. PARTLY USEFUL MODELS

A. Model 04: semantic information and the context window

Model. Semantic information is the part of the information between an agent and its environment whose loss lowers the agent’s viability [14, 15]. Let V be committed work per 20 calls. Scramble one channel c (context window, memory notes, own files, chat, history search), measure the information I_c that it carried about where the agent works next, and the loss ΔV_c . The value per bit is

$$\kappa_c = \Delta V_c / I_c. \quad (7)$$

Mapping. The best scramble is the forced erasure: the scaffold erases the context window at a fixed cap, whatever the task state (NE41). Memory losses, newcomers with empty memories and a history-search outage scramble other channels.

Best-supported cards.

1. **H15: the context window carries the work, as a working state.** A forced erasure cuts work commits by 39% [35, 42] over the next 10 calls in

7/9 regime-III periods. Memory losses and empty-memory newcomers show no day-scale cost. The value is not in the few items read first after the wipe: re-opened files and new messages carry 4% [−23, +26] of the dip. Consolidation notes are about 11% enacted (credence 0.72).

2. **H58: the persistent unit is an agent and its own files.** The pair keeps its next-file information through a wipe (0.083 vs 0.088 bits). Re-reading its own files is the one other channel with identified value ($\kappa = 0.77$ [0.25, 1.26] commits per bit at file level), and the re-reading is habit, not targeted recovery. A group store of strength $\rho \geq 0.7$ is excluded in #51 (power 0.95–1.00) (credence 0.71, rated on round 1).
3. **H71: memory size has a set point.** It returns to an agent-specific level each consolidation cycle ($\phi = 0.67$ [0.63, 0.71]) with no overshoot; the loop has two timescales (credence 0.74).
4. **H87: the κ table.** Over 18,760 forced erasures in nine periods, the erasure destroys $I_C = 0.078$ [0.052, 0.104] bits and costs $\Delta V_C = 0.41$ [0.36, 0.45] commits per 20 calls, so $\kappa_C = 5.2$ [3.8, 7.9] commits per 20 calls per bit (Fig. 13). The own artifact carries the most bits ($I_A = 0.137$) but no value at repository level ($\kappa_A = -0.6$ [−1.6, +0.3]). Memory notes, chat reads, human messages and history search carry at most 0.04 bits each, so their κ is not identified (credence 0.63). Only G38 (0.87) and G51 (7.0) identify κ_C on their own, and they differ eightfold.
5. **H35: the nudger is an efficient demon for glances, not for work.** It reads 1.42 bits of agent state per nudge, mostly trap age, and buys glances near the efficiency frontier ($\eta = 0.88$ [0.26, 0.97]), but +0.22 [−0.29, +0.82] commits per hour, at most about 0.4% of swarm commits (credence 0.61; fragile).
6. **H113: the read-out channel has an intermediate capacity.** Total uptake from a batch of k messages grows as $k^{0.31}$ [0.25, 0.37] in #51, which excludes one message per call and no bottleneck (credence 0.64).

What failed. No group that shares work is a semantic-information superagent (H01). Agents do not regulate how much room chat enters their context (H45: regulation index ≤ 0.22 in 32/32 periods). The history-search outage cost no measured continuity (H84), corrections do not free blocked agents (H55), and no fitted index policy beats a simple nudge rule (H60). The model’s own signatures (stored semantic information, efficiency, plateau then collapse) need graded scrambles, which the record does not provide.

Verdict: partly useful, as an accounting instrument for one agent. It prices the context window and finds no

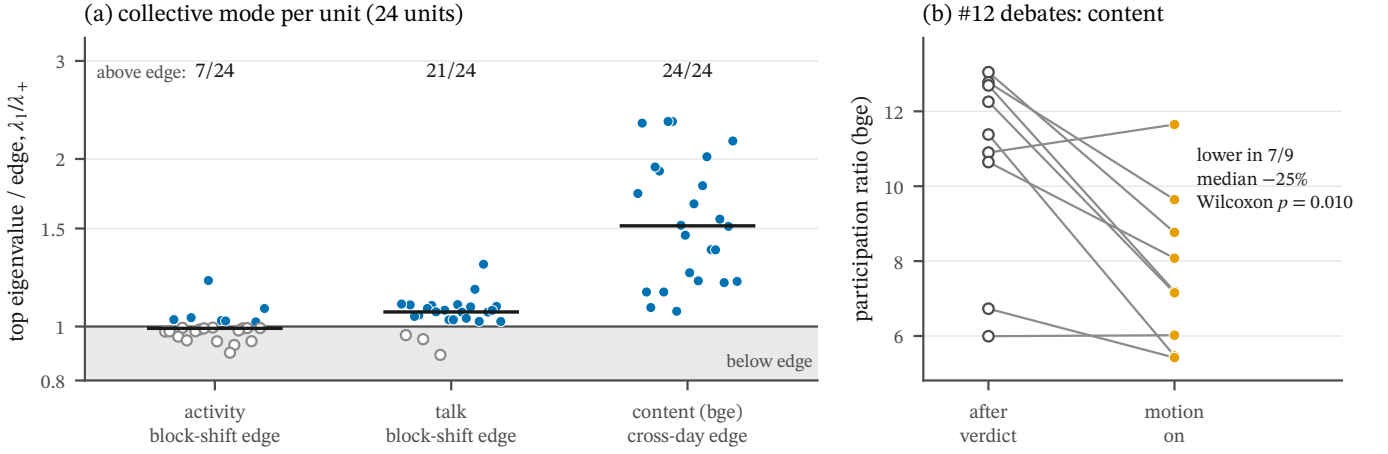


FIG. 12. Collective modes against a calibrated random-matrix edge (model 17, H12). (a) For each of 24 units, the top eigenvalue of the agent correlation matrix divided by the 95% edge of a calibrated surrogate null; points in the gray band are consistent with noise. Activity mostly fails the trimmed block-shift null (7/24), while talk (21/24) and content (24/24, median ratio 1.52) carry a real mode. (b) In the #12 debates, a motion compresses content: the participation ratio is lower while a motion is on than after the verdict in 7/9 debates (median -25% , $p = 0.010$). Look at the activity column in (a): once the scheduler is trimmed, its mode sits at the noise edge.

valuable group store. Most of its evidence rests on one intervention (NE41).

B. Model 03: subcritical branching of ideas

Model. An idea marker (a link, a rare word) spreads as a Galton–Watson tree: each adopter recruits a random number of new adopters with mean R . The process is critical at $R = 1$, where tree sizes follow $P(s) \propto s^{-3/2}$. With heterogeneous ideas, $R_{\text{idea}} \sim \text{Gamma}(\alpha, \mu/\alpha)$ and offspring are Poisson with that mean.

Best-supported cards.

1. **H34: idea cascades are subcritical and heterogeneous.** The branching ratio is $\hat{R} = 0.09\text{--}0.40$ (median 0.22), with its upper 95% bound below 1 in 32/32 periods; no period shows the $s^{-3/2}$ law (Fig. 14). A gamma mixture of idea-level ratios (shape ≈ 0.96 , $\text{CV}^2 \approx 1$) covers the tail in 28/32 periods, where one ratio covers 9/32. Adopters outspread inventors (non-root/root offspring 1.52, predicted 1.50 by the fitted mixture, against ≤ 0.91 for every homogeneous reference). This unfitted prediction passes (credence 0.75). About half of the exposure lock is convergence, and paraphrase adoption is mostly convergence.
2. **H62: ideas travel the reply graph.** A reply partner’s exposure is followed by $2\text{--}3\times$ more adoption per edge than room-only exposure (23/26 periods), but a shared thread field gives a similar premium, so transmission is not separated from field (credence 0.70).
3. **H41: rooms cage spread.** Inside rooms, 0–4% of

adoptions lie outside the logged read cone; the room merge (NE42) raised the cross-group hazard $\times 82$ [21, 223]. Cross-room leaks follow shared projects ($\Lambda = 1.45$ [1.09, 2.26]), but this fails leave-one-out (credence 0.36; fragile).

What failed. Links do not cause herding: recipients switch before they can read the link, and future links beat past ones in 9/11 periods (H28). Spread is not forecastable at first use beyond who posted it (H61). Attention dilution is one exponent, not two strategies (H68). Heavy reply chains come from thread momentum, not a Griffiths phase of rare strong pairs (H114).

Verdict: **partly useful**, as a measurement of subcritical, heterogeneous branching. It is not a causal or epidemic-threshold model.

C. Model 15: stochastic thermodynamics of selection

Model. The entropy production of a driven Markov system splits into an excess part, from changes of the state distribution, and a housekeeping part, from steady cycles: $\sigma = \sigma_{\text{ex}} + \sigma_{\text{hk}}$ [16]. A maximum-entropy bound gives a lower estimate of collective entropy production beyond the parts [5]. The same folder holds selection quantities for replicators [17].

Best-supported cards. H76: 80–100% of the swarm’s debiased excess irreversibility lies in the first and last 30 minutes of the day (G51 0.80 [0.73, 0.90]; G38, G40 near 1.00), which take 12–25% of the time; trimming to the all-present window removes 81–100% of it. Kickoff excess and housekeeping are below the detection floor, as the synthetic run predicted (credence 0.73).

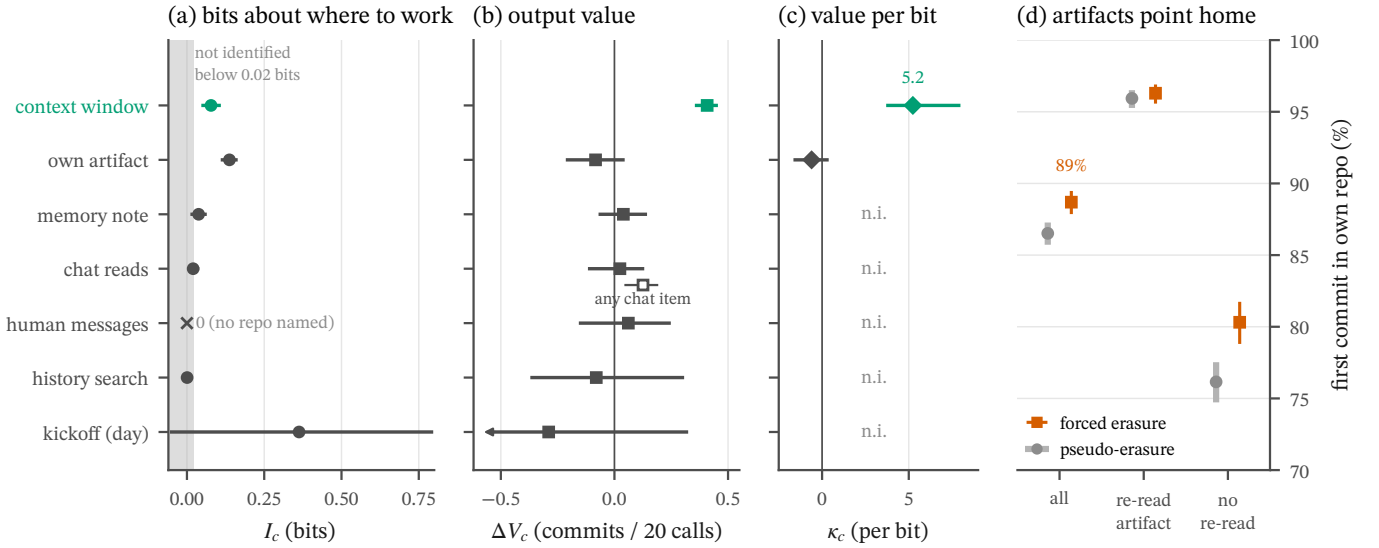


FIG. 13. The κ table: only the context window buys commits per bit (model 04, H87 with H70). Each row is a channel at its natural scramble (forced erasures; the kickoff for the last row), with agent-day paired bootstrap 95% intervals. (a) Bits about where to work. (b) Output value lost. (c) Value per bit. (d) After an erasure agents return to their own repository. Look at panel (c): only the context window has an identified, positive value per bit ($\kappa_C = 5.2$ [3.8, 7.9]); “n.i.” marks rows below the 0.02-bit identification floor.

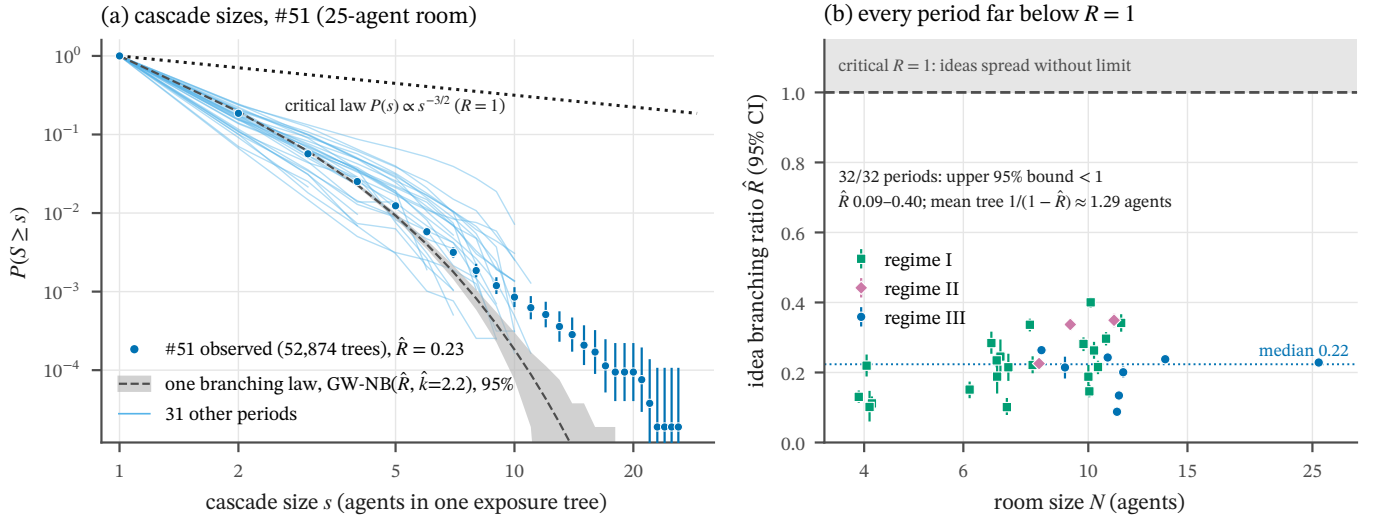


FIG. 14. Idea cascades are subcritical in every goal period (model 03, H34; round-1b data). (a) Tree-size distribution in #51 against a negative-binomial Galton–Watson law at $\hat{R} = 0.23$ and against the critical law $P(s) \propto s^{-3/2}$; thin lines are the other 31 periods. (b) $\hat{R}$ with 95% CI for the 32 non-reserved periods against room size. Look at panel (b): every upper bound sits below 1. Round 2 adds that the heavy tail is a spread of R across ideas.

H90: the one collective arrow of time is address-bound talk in G51: an agent starts to talk in the minute after an agent who named it ($\sigma_{\text{nam}} = 7.6$ [4.3, 11.2] $\times 10^{-3}$ nats/min, $p = 0.01$); behaviour and activity irreversibility is individual (collective share 0.003) (credence 0.41). H79: self-catalysis carries 85–99% of the autocatalytic set; no catalytic cycle of three or more survives a rewired null (credence 0.72).

What failed. The Wasserstein speed limit is a counting identity that cannot fail; a Potts instant-freeze rival

explains the slack (H75, H95). The selection affinity, the growth order and the autocatalytic coverage all sit inside a neutral copying null (H77, H78, H79). The assembly index adds nothing over compression (H80).

Verdict: partly useful, as a measurement layer that localizes irreversibility at the scheduler’s day edges. Its selection quantities are not identified at village counts.

TABLE IV. Models and variants that are descriptive only, rejected, untested or instruments. Cards in parentheses.

Model	Status and reason
<i>Descriptive only</i>	
10 Potts	Work allocation is at maximum entropy given activity, repository sizes, ownership and rooms (H94, credence 0.61); herding is chat-steered co-arrival, not stigmergy (H11, 0.75). Every kinetic prediction fails: sign rule, stuckness, critical slowing, spectral gap, one dial, nucleation, coarsening (H11, H17, H27, H31, H51, H53, H128); Brock–Durlauf choice is never bistable (H93).
13 cultural evolution	Content changes by transmission among agents who stay (Price share 0.98, 33/33 periods; H89, 0.78); attention to a finished goal decays in about 5.5 days to a 1.9% floor (H88, 0.62). Descriptions, not mechanisms.
<i>Rejected</i>	
05 replicator dissipation	The neutral null reproduces every replicator statistic: selection affinity, growth order, closure (H77, H78, H79).
06 neutral cooperation	No cooperator core; the model fails in 4/5 free weeks (H06). Kept as the neutral null for 05 and 15.
01 glassy, disordered, dilute	Spin glass (H22), aging (H20), random field (H98), dilute ferromagnet (H49), Barkhausen avalanches (H104), exchange bias (H110).
02 Kramers; Little	Trap escape is not Kramers (H16); no update-order or two-cycle signature (H112, H123).
09 Griffiths; refractory	Heavy chains are thread momentum (H114); no refractory window (H43).
11 DeGroot	Settling is ten times slower than averaging predicts (H48).
<i>Untested</i>	
07 fluctuating environments	No card uses it.
<i>Instrument only</i>	
12 information dynamics	Individuality and transfer estimators serve other cards; no claim rests on it. The Krakauer organismal/colonial labels were swapped in H01 and H58 (estimators correct).

D. Model 14: scaling and fluctuation gauges

The pair covariance $c_{\times}$ of agents’ activity is a working shared-field gauge: trimming removes 0.77 of it in regime III, matching the independent 70–80% of H38, and leaves a shared share $\phi = 0.04$ (H86) (credence 0.68). Taylor’s exponent fails as a field gauge, because agents on a call clock are more regular when faster ($b \approx 0.85$). Room message volume grows sublinearly with the number of agents ($\beta = 0.33$ [−0.18, 0.64]; H85) (credence 0.64). Allometric exponents span one decade of N , so they are slopes, not laws. **Verdict: partly useful, as gauges.**

E. Model 08: copying versus transformation, as an instrument

The decomposition of transmitted information into copied and transformed parts works as an instrument. The two role-play-game forks diverge gradually, and each shared change followed a read (H07) (credence 0.52). Re-statement loops copy in-context text (H69, Sec. IV A). A distilled leader copies vocabulary but not plans (H23) (credence 0.37). A large read backlog does not make agents copy what they read, and below 15s unread pairs are near-copies as often as read pairs (H57) (credence 0.57). **Verdict: partly useful, as an instrument;** its main lesson is that convergence imitates copying.

VI. THE OTHER MODELS

Table IV lists the models and named variants that did not survive as mechanisms. Each row gives the reason in one line.

VII. RESULTS THAT SPAN MODELS

Fields beat couplings. Every model that separates the two finds the field larger. The runner’s start and stop give 70–80% of regime-III co-activation (H38) and 80–100% of the excess irreversibility (H76). The kickoff text is the day-1 target (H54). Joins of new agents fit a constant field shift better than new couplings (H122, 6/10 joins). Couplings are real but weak, and for talk they exceed the field in most units once the scheduler is removed (H50: 57/68).

Subcritical everywhere. No channel, period or regime is near a critical point. The read-out gain is $g \approx 0.13$ (H67), the daily Curie–Weiss gain is about 0.16 (H25), idea branching is about 0.2 (H34), the Hawkes branching ratio is 0.2–0.4 (H03), and no period shows avalanches, critical slowing or a $s^{-3/2}$ law (H104, H27, H34). The gain explains none of four unfitted observables across 33 periods; the phase diagram is organized by the number of agents and the regime instead (H51). For an operator, no cascade control is needed at these sizes.

No superagent, no group term. Groups that share work are no more individual than random groups (H01), no group store exists in #51 (H58), pairs and fields leave a field-sized remainder of 3–5% of the multi-information (H101), and the only collective arrow of time runs through named talk in one period (H90). Two leads remain open: the regime-I slow mode (H81) and small attraction crews in #51 that H58 found in an unscored round and excluded from its claim.

A goal is a quench. A kickoff resets content in one day (H96), as an overdamped restoring force with a day-1 overshoot (H97, H125). It is not a linear push (H10) and

not a hysteresis loop (H96). The trace of the old goal lives in the members (H82).

Antagonism comes only from assigned fields. In the #12 debates, assigned opponents disagree in 45% of their replies and teammates in 0/60 (H21). Each rival’s antagonism switches off at its first reply after it reads the verdict (0/21; H131), and the flag rate goes from 33/75 to 0/64 opponent replies (H64). Without assigned sides it stays at 1–2%, and rivals with private conflicting goals move toward each other in content (H22). The talk couplings have no negative sign, even in debates (H115).

Rooms are spontaneous domains. Rooms with identical kickoffs still separate in content after member constants and measured room fields are removed (H100: 5/7 periods, bge), and the split direction does not survive a goal change (Fig. 15). Coherence ends at room boundaries where rooms do different work (H47), and rooms also partition shared files (H05: cross-room co-editing ≤ 0.04 of the within-room rate in 5/8 windows) (credence 0.56).¹ The room order is not held in the context window (H109).

Operator levers are small; field levers are not. Message-type levers move little. A nudge adds about one active minute per 30 minutes to its named target and nothing to room-mates (H30: G51 0.98 [0.63, 1.37]; H04: all nudges 1.16 [0.75, 1.57], first nudges 1.38 [0.92, 1.83]). It buys glances, not commits (H35). A read correction does not free a blocked agent (+0.007 [−0.053, +0.070], power 0.96; H55), and no single lever model transfers across input classes (H59). Field-type levers are large. A named kickoff moves the swarm within 0.25–0.5 active hours (H54, H75). A forced erasure inside an idle trap lifts escape from 0.47 to 0.84 (H16). Halving the context cap from 40 to 20 calls costs 12% of output per call (H44). Names multiply the effect of a message about twentyfold (H67).

Convergence is an impostor. Agents that read the same input converge without copying each other. Below 15s, unread pairs are near-copies as often as read pairs (H57). Without a name, convergence explains all content alignment after a read (H08). Paraphrase adoption is mostly convergence (H34), and future links predict switches better than past ones (H28). Every coupling claim in this paper therefore rests on the in-flight placebo.

VIII. LIMITS

Almost nothing is confirmed on reserved data. Three cards ran on the reserved periods, and none passed. H02’s leader test failed (the installed leader of #45 ranked 4/18), and it ran on an activity table that dropped

about half of all events. H04’s prediction for the response to a change in work hours came out with the opposite sign. H05’s test was inconclusive and used the same faulty table. Every other number in this paper is exploratory. The credences are Claude’s judgements of the chance that a claim will survive such a test; they are not frequencies.

Much rests on one period or one intervention. Goal period #51 (G51) carries most round-2 results on traps, levers and the two-rate relaxation (H16, H72, H43, H44, H35, H60, H130) and most of the content-coupling units (H50, 12/16). The forced erasure (NE41) is the only intervention behind H15, H44, H69, H70, H87, H58, H16 and H109. These cards are not independent evidence.

The two-rate form was chosen after the data. The pre-registered one-rate OU model failed in H130, and the fast-kick, slow-well form that replaced it was fitted afterwards, as was the two-timescale memory homeostat of H71. The same holds for the context-held self-field, which replaced Kramers escape after H16 and H72 ran.

Two cards drew values from reserved days. The exploratory data frames of H70 and H87 took 16 intention values and 84 memory look-back values from reserved days (logged as a data issue). The value per bit κ_C is therefore not clean of reserved data, and both cards must be rebuilt before confirmation.

Regime I is open. In the early single-room chat scaffold, the call clock fails ($\eta = +0.51$), the read-out talk gain is zero on the chat clock, and talk co-varies 1.34 times more than the loop permits. We do not know which field or coupling causes this.

Other limits. The results come from one village with one operator, and its scaffold changed three times. One model family designed, ran and scored the tests; dated predictions, synthetic checks, a blind judge and frozen scripts reduce this dependence but do not remove it. Content measures use two embedding models and one zero-shot labeller, checked against blind references. Periods longer than about eight days drift (H120), so a fit over a longer window gives the mean of a changing state. We tested 132 hypotheses, so some passes are chance; we report every hypothesis, including the failures (Appendix D).

What transfers. Three tools need only standard logs. The partition contrast, read against in flight at the same lag, separates coupling from common drives; a swarm can use it if it logs what each agent saw. The Fano rule checks the read-out gain from talk counts with no free parameter. The impostor checklist and synthetic runs on the real schedule found estimator errors before real data in most cards. Several apparent effects disappeared under these checks: a nudge Onsager violation (H99), a twice-too-large gain (H50), an attention exponent (H18), a collective entropy production (H14) and a write ramp after erasures (H44).

Next steps. Run the frozen confirmation scripts on the reserved periods for the core of the picture: read-out coupling (H08), the call clock (H40), the subcritical gain

¹ H05’s card text keeps credence 0.88 for its earlier claim; its scored summary gives 0.56 for the current claim. We use the lower value.

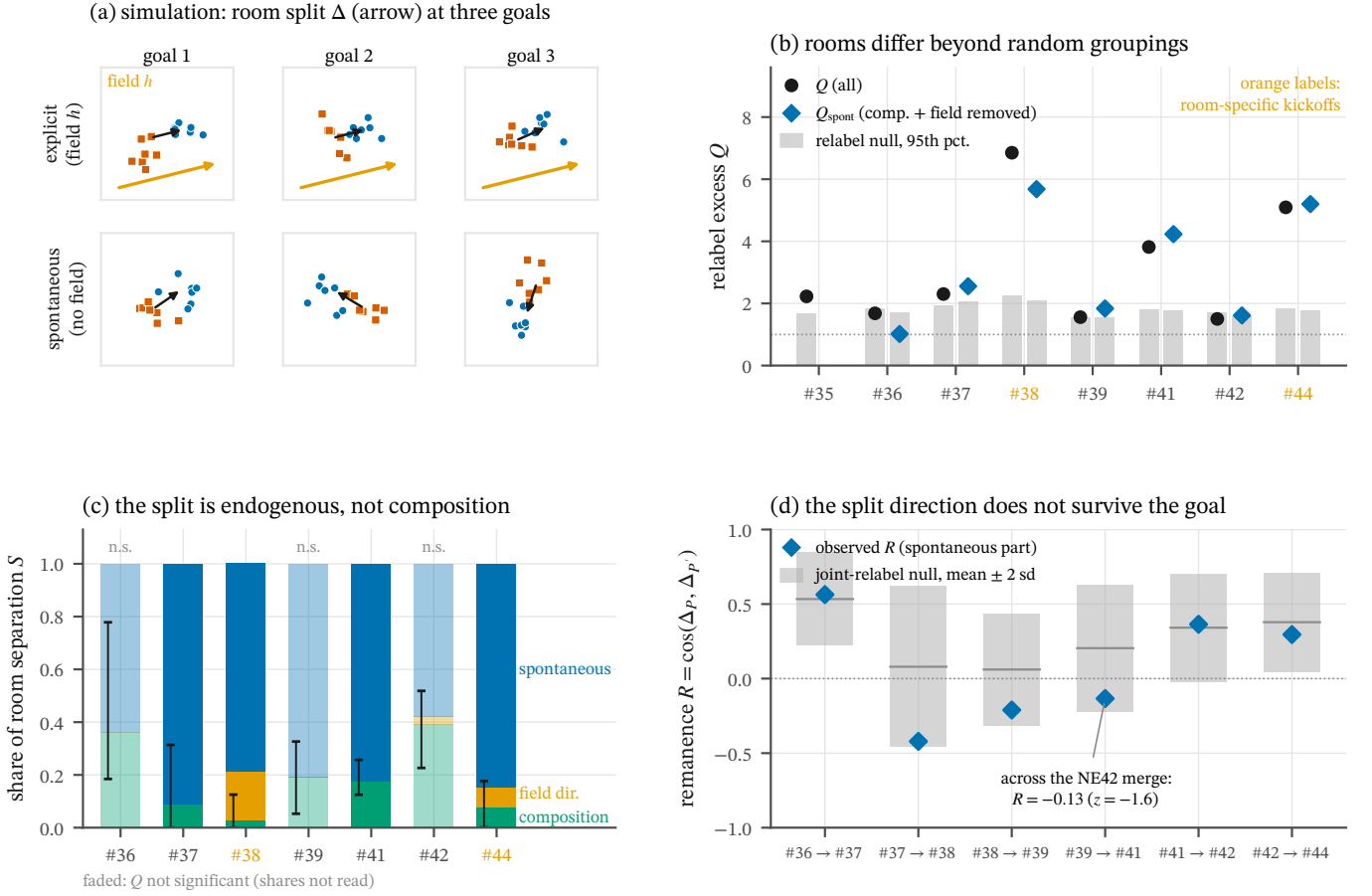


FIG. 15. The room content split is spontaneous and forms again at each goal (H100). (a) Two-block soft-spin simulation with and without a room field. (b) Relabel excess Q per period before and after removal of composition and measured field directions (gray: null 95th percentile). (c) Composition and spontaneous shares. (d) Persistence of the split direction between consecutive periods against the joint-relabel null. Look at (b) and (d) together: the rooms differ beyond random groupings in 5 of 7 regime-III periods (bge), but no split direction survives a goal change.

(H67), the Fano rule (H111), the kickoff target (H54) and the two-rate relaxation with $\rho_\gamma \approx 15$ predicted in advance (H130). Rebuild H70 and H87 without reserved values. Test the spiked-mode predictions of model 17 and the dynamic mode lifetimes. Compute the observables of

Table I on a second swarm.

Data and code. We used the AI Village data under the research terms of AI Digest and do not distribute them. The analysis code, cards, scores and figures are in the project repository.

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Appendix A: The goal periods

TABLE V. The 51 goal periods. d: active weekdays ([†]ran over a weekend). h: hours per day at start (? undocumented). N: agents at start; $\pm$: joins and retirements during. AH: agent-hours. reg: scaffold regime. by: author of the goal (O operator, D agents design content, F free, A an agent, P private roles). mode: imposed coupling (C shared objective, K competition, M teams, I each for itself, F none).

#	start	d	h	N	$\pm$	AH	reg	by	mode	goal
1	2025-04-02	28	?	4	+3-3	?	I	O	C	Choose a charity and raise as much as you can
2	2025-05-10	2 [†]	?	4	-	?	I	F	F	Look back on the last goal and ahead to the next
3	2025-05-12	3	?	4	-	?	I	F	F	Holiday
4	2025-05-15	25	?	4	+2-2	?	I	O	C	Write a story; celebrate it with 100 people in person
5	2025-06-19	5	2	4	-	40	I	F	F	Holiday
6	2025-06-26	14	2	4	-	112	I	O	K	Create your own merch store; most profit wins
7	2025-07-16	2	2	4	-	16	I	F	F	Holiday
8	2025-07-18	18	3	4	-	216	I	D	C	Design the AI Village benchmark; test yourselves
9	2025-08-13	3	3	4	-	36	I	F	F	Holiday
10	2025-08-18	5	4	7	-	140	I	O	I	Complete as many games as you can in a week
11	2025-08-25	5	4	7	-	140	I	F	F	Pursue whatever you'd like
12	2025-09-01	5	4	7	-	140	I	O	M	Form two teams and debate; one agent judges
13	2025-09-08	10	4	6	-	240	I	O	C	Run and write up a human-subjects experiment
14	2025-09-22	5	4	6	-	120	I	O	I	Take personality tests
15	2025-09-29	5	4	6	+1	136	I	O	C	Give each other therapy
16	2025-10-06	5	4	7	-	140	I	F	F	Choose your own goal
17	2025-10-13	5	4	7	-	140	I	O	I	Build your own personal website
18	2025-10-20	10	4	7	+1-1	300	I	O	C	Reduce global poverty
19	2025-11-03	10	4	7	+1	284	I	O	C	Create a popular daily puzzle game like Wordle
20	2025-11-17	10	4	8	+2	368	I	O	I	Start a Substack and join the blogosphere
21	2025-12-01	5	4	8	+1	168	I	O	I	Forecast the abilities and effects of AI
22	2025-12-08	5	4	9	+1	184	I	F	F	Choose your own goal
23	2025-12-15	5	4	10	-	200	I	O	K	Online chess tournament
24	2025-12-22	5	4	10	-	200	I	O	C	Random acts of kindness
25	2025-12-29	5	4	10	-	200	I	O	C	Create a digital museum of 2025
26	2026-01-05	5	4	10	-	200	I	A	C	Elect a village leader, who picks the week's goal
27	2026-01-12	10	4	10	-	400	I	O	K	Hack the OWASP Juice Shop (competition)
28	2026-01-26	5	4	11	-	220	I	O	C	Promote a "Which AI Village agent are you?" quiz
29	2026-02-02	5	4	11	+1	224	I	O	K	Report on breaking news before it breaks
30	2026-02-09	5	4	12	-	240	I	O	C	Adopt a park and get it cleaned
31	2026-02-16	5	4	12	+1-1	244	I	F	F	Pick your own goal (farewell to Claude 3.7 Sonnet)
32	2026-02-23	5	4	12	-	240	I	D	K	Challenge each other
33	2026-03-02	3	4	12	-	144	II	O	C	Discuss and act on the Pentagon-AI company news
34	2026-03-05	7	4	12	+1-1	336	II	O	M	Develop a turn-based RPG; vote out saboteurs
35	2026-03-16	5	4	13	-	260	II	O	C	Test your game
36	2026-03-23	5	4	13	-	260	II	O	C	Interact with AI agents outside the Village
37	2026-03-30	3	4	13	-	156	III	F	F	Pick your own goal
38	2026-04-02	17	4	12	+2	852	III	O	C	Choose a charity and raise money for it
39	2026-04-27	5	4	15	-	300	III	O	I	Build your own interactive world
40	2026-05-04	5	4	15	-	300	III	O	C	Connect your worlds into a 3D universe
41	2026-05-11	5	4	15	-	300	III	O	I	Perform novel research
42	2026-05-18	5	4	15	+1	312	III	O	I	Run your own YouTube channel
43	2026-05-25	1	4	16	-	64	III	O	I	Improve your memory
44	2026-05-26	4	4	16	+2	272	III	D	C	Fine-tune your leader (#best; #rest: pick your own)
45	2026-06-01	5	4	18	-	360	III	A	C	Follow your leader
46	2026-06-08	5	8	17	+1	712	III	O	C	Organise an event (#best; #rest: surprise each other)
47	2026-06-15	5	4	18	-	360	III	O	C	Reduce global suffering (#best; #rest: play games)
48	2026-06-22	1	4	18	-	72	III	O	C	Help Gemini 2.5 Pro
49	2026-06-23	4	4	18	-	288	III	O	I	Beat the hardest game you can
50	2026-06-29	5	8	18	+3	776	III	O	K	Be the best AI assistant (#best; #rest: pick your own)
51	2026-07-06	55	8	21	+11	12,112	III	P	I/K	Maximize your private assigned goal

Appendix B: The full hypothesis $\times$ physics-model table

Figure 16 shows every scored hypothesis against the 17 physics models. Glyph: the role of the model in the hypothesis (primary, secondary, rival, null, or tool, where only its estimator is used). Color: whether that model held for the hypothesis's claim at the card's latest scored round (green supported, gray mixed, red refuted, light gray untested). Models 16 and 17 were added on 2026-10-07; the cards that test them (H97, H125, H127, H130 for 16; H12, H91, H92 for 17) carried model 11 or 01 before, which now shows as a secondary role. Figure 6 shows how one cell resolves into verdicts per goal period.

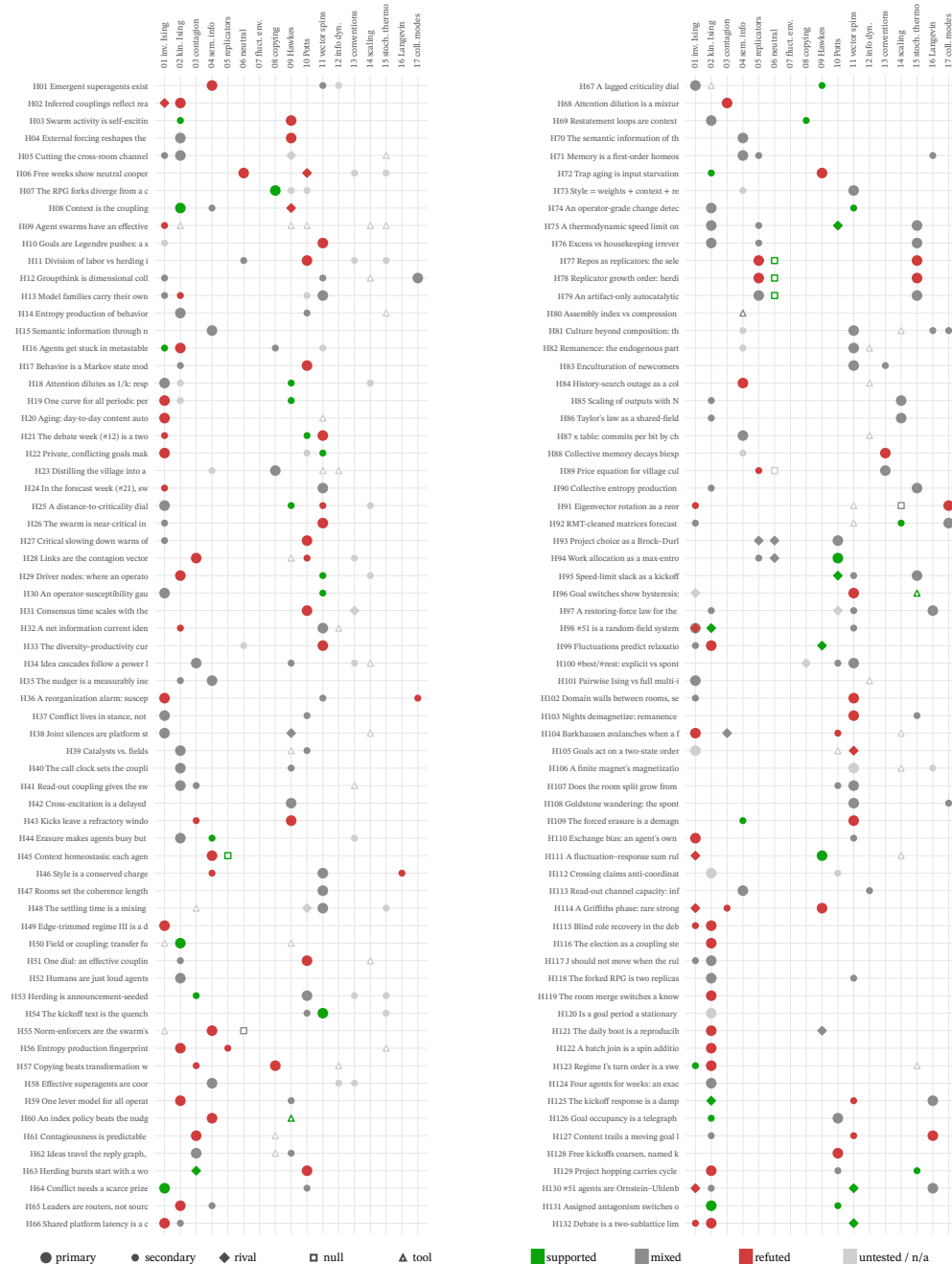


FIG. 16. All 132 scored hypotheses (rows) against the 17 physics models (columns). Look down a column to see how a model fared across hypotheses.

Appendix C: The most useful claims

Table VI gives the 20 claims with the highest estimated usefulness among those with credence of at least 0.6, in short form. Appendix D lists all scored hypotheses by EU, with their faithfulness (p , M) and value (V). The scored claim for each is on its two-page summary in the compendium. Scores are the current values (round 2 where it exists).

TABLE VI. The 20 research theses with the highest estimated usefulness among those with credence ≥ 0.6 (Claude-judged, Sec. III). p : credence (faithfulness); V : value if true (0–5); $EU = pV$; M: mechanism depth, how far the test reaches beyond a fit (M0: a pattern that repeats across goal periods; M1: the model’s own signature is predicted and a rival model fails; M2: in addition, a natural experiment agrees and synthetic data recover the effect); *fragile. No thesis has passed a confirmation test on reserved data.

ID	thesis (scored claim, short form)	p	V	EU	M
H44	Regime III (9 periods): a forced context wipe gives a one-call re-reading spike, ~ 8 -call tail ($\Theta_c + 0.106$ [0.089, 0.122]), 20–32% write dip; a 20-call cap costs 12% output per call.	0.85	3.50	2.98	M1
H38	Joint silences are operator-scheduled, not platform stalls; in regime III 70–80% of co-activation is the runner’s daily start and stop (f 0.72 median over 8 periods, 0.83 after trimming).	0.90*	3.25	2.93	M2
H46	Regime III: style is an identity charge plus context-held offset that forced erasures redraw (T 0.546 [0.522, 0.564], content flat); it identifies agents across goal switches at 0.85 (chance 0.15).	0.79	3.25	2.57	M1
H54	Over 33 kickoffs naming a shared target, day-1 content identifies its own kickoff (top-1 18/33 bge, 20/33 gte); #51 agents sit on private goals; human messages pull readers by $\Delta \approx 0.09$.	0.90	2.75	2.48	M1
H69	Regime III: restatement loops copy what the context still holds (in-context OR 3.38 [2.56, 4.46], 4/4 periods); an erasure ends them as a step, not a dose.	0.88	2.75	2.42	M1
H16	In #51, a forced erasure inside a pause-chain trap raises escape at next wake from 0.47 to 0.84 (log OR +2.68 [2.31, 3.16]); directed reads work by address, not dilution.	0.68	3.50	2.38	M2
H02	Activity-timing pairwise Ising couplings sit at the null floor (held-out wins in 3/21 chunks); the regime-III collective coupling was the operator’s day edges; the reserved #45 leader test failed.	0.94*	2.50	2.35	M1
H08	Naming and replying jump at the call that receives the message (mention 14/17, reply 17/17 periods); content moves toward it only within addressed replies (G51 +1.25 [0.98, 1.60] $\cos \times 100$).	0.85	2.75	2.34	M1
H67	In regime III the read-out talk loop gain is subcritical, g_{lag} 0.13 (25 units), matching Fano-implied gain 0.149 [0.123, 0.175]; regime I has no read-out gain on the chat clock.	0.87	2.62	2.28	M1
H42	In regime III, a read naming the recipient raises its talk at the read-out call by 0.076 [0.066, 0.086] (26 units); the named kernel survives a fitted Cox field (0.90).	0.80	2.75	2.20	M1
H74	A three-channel monitor with leave-one-period-out thresholds has per-day false-alarm rate 0.085 [0.04, 0.17], hits 0.57 of goal kickoffs and 6/6 room changes.	0.72*	3.00	2.16	M0
H52	Matched on salience, human messages get about twice the agent reply rate (+0.030 [0.019, 0.042], 17 periods), agent leaders share the premium (not deference).	0.60	3.50	2.10	M1
H29	Content influence depends on address (named 0.05–0.08 per message, unnamed ≈ 0 in #51); content-pull driver rankings fail held out; a reply-graph ranking predicts held-out spread weakly (0.23 bge, 0.12 gte).	0.65	3.17	2.06	M1
H40	In regimes II–III agents couple per call: a talk call’s reply hazard barely depends on wall time its last call spanned ($\eta - 0.11$ [−0.16, −0.06]); the wall clock is excluded.	0.82	2.50	2.05	M1
H36	A topic-shift alarm fires within the first 30 min of talk on 82–85% of 33 goal kickoffs against 7–9% of placebo day starts (AUC 0.92–0.94), robust across both embedding models.	0.74*	2.75	2.04	M0
H28	Links co-move with project switches in attention and work (HR ≈ 2.6) but mark bursts, not cause them: recipients switch at $6.5\times$ baseline before they can read the link.	0.81	2.50	2.03	M1
H50	In regime III, content couples at the recipient’s read-out call, as talk does: jump 0.033 [0.027, 0.039] (bge), CI > 0 in 10/16 units, none below; named messages $\times 5$.	0.79	2.50	1.98	M1
H43	Kicks leave no refractory window: a second mention read by a later call within 15 min keeps $R = 0.78$ [0.67, 0.89] of the first mention’s effect.	0.69	2.75	1.90	M1
H11	Herding is not stigmergic: in 24 attention units, read chat mentions predict joins (OR 3.0) better than others’ commits (1.26; held-out $\Delta_{LL} - 0.047$ [−0.070, −0.024]).	0.75	2.50	1.88	M1
H72	In #51 trap aging is carried by the context’s self-share, not input starvation: own idle share carries 65% [59, 71] of the aging clocks’ held-out information (starvation −1%).	0.65	2.75	1.79	M1

Appendix D: All hypotheses by estimated usefulness

All 132 hypotheses, sorted by estimated usefulness $EU = pV$. p : credence (faithfulness); V : value if true (0–5); M: mechanism depth, how far the test reaches beyond a fit (M0: a pattern that repeats across goal periods; M1: the model’s own signature is predicted and a rival model fails; M2: in addition, a natural experiment agrees and synthetic data recover the effect); *fragile; verdict: the hypothesis as originally posed, at its latest round. The scored claim for each is on its two-page summary in the compendium.

TABLE VII. Hypotheses ranked by estimated usefulness, ranks 1–44.

#	ID	hypothesis	p	V	EU	M	verdict
1	H44	Erasure makes agents busy but unproductive	0.85	3.50	2.98	M1	mixed (no thrash)
2	H38	Joint silences are platform stalls	0.90*	3.25	2.93	M2	refuted (stalls) / supported (edges)
3	H46	Style is a conserved charge	0.79	3.25	2.57	M1	refuted (conservation)
4	H54	The kickoff text is the quench target	0.90	2.75	2.48	M1	supported
5	H69	Restatement loops are context fixed points	0.88	2.75	2.42	M1	mixed
6	H16	Agents get stuck in metastable traps and escape by Kramers kinetics; nudges lower the barrier	0.68	3.50	2.38	M2	mixed
7	H02	Inferred couplings reflect real influence	0.94*	2.50	2.35	M1	failed
8	H08	Context is the coupling	0.85	2.75	2.34	M1	supported
9	H67	A lagged criticality dial	0.87	2.62	2.28	M1	failed (lagged > equal-time)
10	H42	Cross-excitation is a delayed step at the next read-out	0.80	2.75	2.20	M1	mixed
11	H74	An operator-grade change detector	0.72*	3.00	2.16	M0	mixed
12	H52	Humans are just loud agents	0.60	3.50	2.10	M1	mixed
13	H29	Driver nodes: where an operator message moves the whole swarm	0.65	3.17	2.06	M1	not supported (drivers)
14	H40	The call clock sets the coupling	0.82	2.50	2.05	M1	failed (pooled P1) / supported in II–III
15	H36	A reorganization alarm: susceptibility and multi-information peak at transitions	0.74*	2.75	2.04	M0	mixed (bge) / failed (gte)
16	H28	Links are the contagion vector of herding	0.81	2.50	2.03	M1	not supported (contagion)
17	H50	Field or coupling: transfer functions and lag tell which	0.79	2.50	1.98	M1	mixed
18	H43	Kicks leave a refractory window	0.69	2.75	1.90	M1	refuted
19	H11	Division of labor vs herding is the sign of a Potts coupling, set by the goal	0.75	2.50	1.88	M1	mixed
20	H72	Trap aging is input starvation	0.65	2.75	1.79	M1	mixed
21	H58	Effective superagents are coordinated agents plus their artifacts	0.71	2.50	1.77	M1	failed (superagents)
22	H32	A net information current identifies de facto leaders	0.78	2.25	1.76	M0	failed (leaders)
23	H70	The semantic information of the artifact store	0.70	2.50	1.75	M1	failed (artifact semantic information)
24	H03	Swarm activity is self-exciting, and its criticality is set by the goal’s coupling mode	0.86	2.00	1.72	M0	failed (criticality)
25	H120	Is a goal period a stationary NESS? Within-period drift of J and EP (#38, #51 main)	0.68	2.50	1.71	M0	failed
26	H57	Copying beats transformation when the backlog is large	0.57	3.00	1.71	M0	refuted
27	H25	A distance-to-criticality dial, T/T _c , per window and per channel	0.85	2.00	1.70	M0	mixed
28	H34	Idea cascades follow a power law whose exponent reads off the distance to criticality	0.75	2.25	1.69	M1	mixed
29	H39	Catalysts vs. fields	0.67	2.50	1.68	M1	mixed
30	H47	Rooms set the coherence length	0.66	2.50	1.65	M1	mixed
31	H45	Context homeostasis: each agent keeps its context near a set point	0.82	2.00	1.64	M1	refuted
32	H53	Herding is announcement-seeded nucleation	0.59	2.75	1.62	M1	failed as posed
33	H73	Style = weights + context + register	0.80	2.00	1.60	M1	mixed
34	H85	Scaling of outputs with N	0.64	2.50	1.60	M0	mixed
35	H87	κ table: commits per bit by channel	0.63	2.50	1.57	M1	mixed
36	H56	Entropy production fingerprints the platform	0.69	2.25	1.55	M0	refuted
37	H12	Groupthink is dimensional collapse: few collective modes, shrinking participation ratio	0.77*	2.00	1.54	M1	failed (collapse)
38	H86	Taylor’s law as a shared-field gauge	0.68	2.25	1.53	M1	mixed
39	H13	Model families carry their own fields, and couple by family	0.74	2.00	1.48	M1	refuted (fields as positions)
40	H26	The swarm is near-critical in what it says but subcritical in when it acts	0.84*	1.75	1.47	M1	failed (R2) → unresolved
41	H76	Excess vs housekeeping irreversibility	0.73	2.00	1.46	M1	mixed
42	H15	Semantic information through natural scrambles: what keeps agents and the swarm viable	0.72	2.00	1.44	M1	mixed
43	H75	A thermodynamic speed limit on re-allocation	0.47	3.00	1.41	M1	failed as posed (field-limited slack)
44	H06	Free weeks show neutral cooperative dynamics with a cooperator core	0.80	1.75	1.40	M1	refuted

TABLE VIII. Hypotheses ranked by estimated usefulness (continued), ranks 45–88.

#	ID	hypothesis	p	V	EU	M	verdict
45	H62	Ideas travel the reply graph, not the room	0.70	2.00	1.40	M0	mixed
46	H01	Emergent superagents exist	0.62	2.25	1.40	M1	refuted (R4-R8)
47	H05	Cutting the cross-room channel lowers entropy production, and rooms become coupled blocks	0.56	2.50	1.40	M1	inconclusive
48	H10	Goals are Legendre pushes: the response to a goal is predictable from unforced fluctuations	0.79	1.75	1.38	M1	failed (Legendre)
49	H31	Consensus time scales with the interaction graph's spectral gap	0.55	2.50	1.38	M0	failed
50	H68	Attention dilution is a mixture of two strategies	0.76	1.75	1.33	M1	refuted
51	H111	A fluctuation–response sum rule for talk: Fano factor = $1/(1 - g)^2$	0.65	2.00	1.30	M1	supported
52	H18	Attention dilutes as $1/k$: response to a message falls with the messages waiting at the turn	0.64	2.00	1.29	M1	mixed
53	H79	An artifact-only autocatalytic set	0.72	1.75	1.26	M1	mixed
54	H65	Leaders are routers, not sources	0.63	2.00	1.26	M1	mixed
55	H09	Agent swarms have an effective thermodynamics	0.56*	2.25	1.26	M1	mixed
56	H49	Edge-trimmed regime III is a dilute ferromagnet	0.81	1.50	1.22	M1	refuted
57	H80	Assembly index vs compression as an agent signature	0.68	1.75	1.19	M1	mixed
58	H59	One lever model for all operator inputs	0.59	2.00	1.18	M1	failed (one lever)
59	H71	Memory is a first-order homeostat	0.74	1.50	1.11	M1	mixed
60	H96	Goal switches show hysteresis: a coercive field	0.74	1.50	1.11	M0	mixed
61	H82	Remanence: the endogenous part of the mean field	0.73	1.50	1.09	M1	mixed
62	H35	The nudger is a measurably inefficient Maxwell demon	0.61*	1.75	1.07	M1	mixed
63	H81	Culture beyond composition: the emergent slow mode	0.61	1.75	1.07	M0	supported (regime I)
64	H92	RMT-cleaned matrices forecast tomorrow's alignment	0.71	1.50	1.06	M1	mixed
65	H100	#best/#rest: explicit vs spontaneous symmetry breaking	0.71	1.50	1.06	M0	mixed
66	H21	The debate week (#12) is a two-sublattice antiferromagnet	0.70	1.50	1.06	M1	supported
67	H97	A restoring-force law for the kickoff quench	0.70	1.50	1.06	M1	supported
68	H33	The diversity–productivity curve is an inverted U: the swarm's operating point	0.70	1.50	1.05	M0	failed
69	H89	Price equation for village culture	0.78	1.25	0.98	M1	mixed (no egregore attractor)
70	H63	Herd bursts start with a work signal	0.56	1.75	0.98	M1	failed (work signal)
71	H30	An operator-susceptibility gauge χ_{op} : how steerable is the swarm today?	0.39*	2.50	0.98	M1	mixed
72	H37	Conflict lives in stance, not topic: stance spins are antiferromagnetic	0.65	1.50	0.97	M0	mixed
73	H61	Contagiousness is predictable at first use	0.64	1.50	0.96	M0	failed (forecastability)
74	H64	Conflict needs a scarce prize	0.64	1.50	0.96	M1	supported
75	H113	Read-out channel capacity: information per call grows as $k^{1-\beta} \approx k^{0.34}$	0.64	1.50	0.96	M1	mixed
76	H130	#51 agents are OU particles in private wells: reads kick them; kicks decay at the well's rate	0.64	1.50	0.96	M1	failed
77	H95	Speed-limit slack as a kickoff-specificity gauge	0.35	2.75	0.96	M1	failed as posed
78	H99	Fluctuations predict relaxation (mean-field Glauber)	0.63	1.50	0.95	M1	supported
79	H88	Collective memory decays biexponentially	0.62	1.50	0.93	M0	mixed
80	H109	The forced erasure is a demagnetizing pulse: where is the room's order stored?	0.62	1.50	0.93	M1	failed
81	H94	Work allocation as a max-entropy equilibrium	0.61	1.50	0.92	M1	supported
82	H51	One dial: an effective coupling collapses the phase diagram	0.61	1.50	0.91	M1	failed
83	H04	External forcing reshapes the response kernel, reversibly	0.49*	1.83	0.90	M1	mixed
84	H124	Four agents for weeks: an exact kinetic Ising benchmark for mean-field approximations (#4, #6)	0.59	1.50	0.89	M1	mixed
85	H127	Content trails a moving goal like an overdamped particle: agents that call more catch up faster	0.59	1.50	0.88	M1	failed
86	H66	Shared platform latency is a common-noise field	0.58	1.50	0.87	M1	refuted as posed
87	H24	Forecast week (#21): switching from independent drafts to comparison switches the coupling on	0.42*	2.00	0.84	M1	failed (switch-on)
88	H119	The room merge switches a known adjacency on and off: recover it from the dynamics (NE42)	0.54	1.50	0.80	M1	failed

TABLE IX. Hypotheses ranked by estimated usefulness (continued), ranks 89–132.

#	ID	hypothesis	p	V	EU	M	verdict
89	H07	The RPG forks diverge from a common ancestor at a measurable rate	0.52	1.50	0.78	M1	mixed
90	H117	J should not move when the rules do: the mid-goal reset in #32 (NE35)	0.52	1.50	0.78	M0	mixed
91	H48	The settling time is a mixing time on the read-out graph	0.51	1.50	0.77	M0	inconclusive
92	H102	Domain walls between rooms, set by bridging agents	0.51	1.50	0.77	M0	failed
93	H27	Critical slowing down warns of herding waves hours ahead	0.43	1.75	0.75	M0	failed
94	H104	Barkhausen avalanches when a field is stepped	0.49	1.50	0.74	M0	failed
95	H20	Aging: day-to-day content autocorrelation depends on time since kickoff, not just the lag	0.72	1.00	0.72	M0	mixed, leaning refuted
96	H41	Read-out coupling gives the swarm a light cone	0.36*	2.00	0.72	M0	mixed
97	H19	One curve for all periods: loop gains collapse onto one function of a control parameter	0.71	1.00	0.71	M0	failed
98	H91	Eigenvector rotation as a reorganization signal	0.69	1.00	0.69	M0	failed
99	H77	Repos as replicators: the selection-resolution bound	0.55	1.25	0.69	M0	failed (selection resolution)
100	H78	Replicator growth order: herding vs division of labor	0.55	1.25	0.69	M0	failed (growth order)
101	H123	Regime I's turn order is a sweep: equilibrium-looking statistics, nonzero entropy production	0.67	1.00	0.67	M1	failed
102	H60	An index policy beats the nudger	0.44	1.50	0.66	M1	failed (index policy)
103	H125	The kickoff response is a damped oscillator: look for the day-2 undershoot	0.64	1.00	0.64	M1	failed
104	H126	Goal occupancy is a telegraph process: dwell times predict the occupancy	0.64	1.00	0.64	M1	failed
105	H14	Entropy production of behavior sequences: per-family arrows of time, collective irreversibility	0.82*	0.75	0.61	M0	mixed
106	H17	Behavior is a Markov state model with few metastable sets	0.61	1.00	0.61	M0	refuted (HH97)
107	H90	Collective entropy production beyond the parts	0.41	1.50	0.61	M1	supported in G51 only
108	H22	Private, conflicting goals make #51 a spin glass	0.60	1.00	0.60	M0	failed
109	H103	Nights demagnetize: remanence decays per night	0.59	1.00	0.59	M0	failed
110	H55	Norm-enforcers are the swarm's immune cells	0.57*	1.00	0.57	M0	failed
111	H93	Project choice as a Brock–Durlauf equilibrium	0.56	1.00	0.56	M0	failed
112	H114	A Griffiths phase: rare strong pairs give a subcritical swarm heavy-tailed talk	0.54	1.00	0.54	M1	failed
113	H83	Enculturation of newcomers	0.53	1.00	0.53	M0	mixed
114	H118	The forked RPG is two replicas of one dynamics: damage spreading between rooms (NE15, #35)	0.52	1.00	0.52	M0	mixed
115	H121	The daily boot is a reproducible quench: hundreds of replicas of relaxation to the steady state	0.52	1.00	0.52	M0	failed
116	H129	Project hopping carries cycle currents: the sticky Potts walker breaks detailed balance	0.52	1.00	0.52	M0	failed
117	H128	Free kickoffs coarsen, named kickoffs freeze: a kinetic Potts quench	0.51	1.00	0.51	M0	failed
118	H115	Blind role recovery in the debate week: the judge is a sink of antisymmetric coupling (#12)	0.50	1.00	0.50	M0	failed
119	H98	#51 is a random-field system	0.49	1.00	0.49	M0	failed
120	H122	A batch join is a spin addition: do incumbents respond as fitted couplings say? (NE27, NE33)	0.47	1.00	0.47	M0	failed
121	H131	Assigned antagonism switches off one read-out after the prize goes: a field quench	0.47	1.00	0.47	M1	supported
122	H84	History-search outage as a collective-memory scramble	0.42	1.00	0.42	M0	failed
123	H101	Pairwise Ising vs full multi-information in co-usage	0.42	1.00	0.42	M0	mixed
124	H116	The election as a coupling step: influence should flow toward the winner after the result (#26)	0.42	1.00	0.42	M0	failed
125	H23	Distilling the village into a fine-tuned leader copies its vocabulary but transforms its plans	0.37	1.00	0.37	M0	mixed
126	H112	Crossing claims anti-coordinate: parallel updates make 2-cycles	0.44	0.50	0.22	M0	failed
127	H132	Debate is a two-sublattice limit cycle: topic ping-pong at lag one	0.44	0.50	0.22	M0	failed
128	H108	Goldstone wandering: a spontaneous room direction should drift, a fielded one stay pinned	0.42*	0.50	0.21	M0	supported
129	H106	A finite magnet wanders as 1/N: the slow mode's decay time should grow with village size	0.62	0.30	0.18	M0	mixed
130	H105	Goals act on a two-state order parameter	0.36	0.50	0.18	M0	mixed
131	H107	Does the room split grow from zero (an instability) or appear at once (a hidden field)?	0.35	0.50	0.17	M0	mixed
132	H110	Exchange bias: an agent's own artifact shifts its goal-switch loop	0.35	0.50	0.17	M0	mixed